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Lower Limb Bones & Joints

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Learning Objectives

Bones: general features of the pelvis, femur, tibia, fibula, tarsals, metatarsals and phalanges

Hip joint: type, associated ligaments, movements permitted

Knee joint: type, associated ligaments, movements permitted

Ankle joint: type, associated ligaments, movements permitted

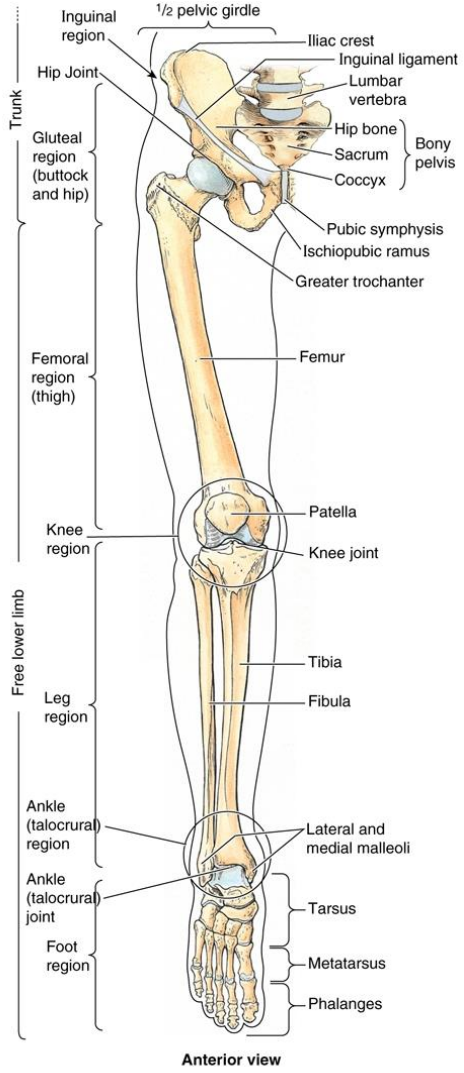
Foot joints: type, associated ligaments, movements permitted

Lower limp regions, bones, and joints

Skeleton of the Lower Limb •

Two separate regions •

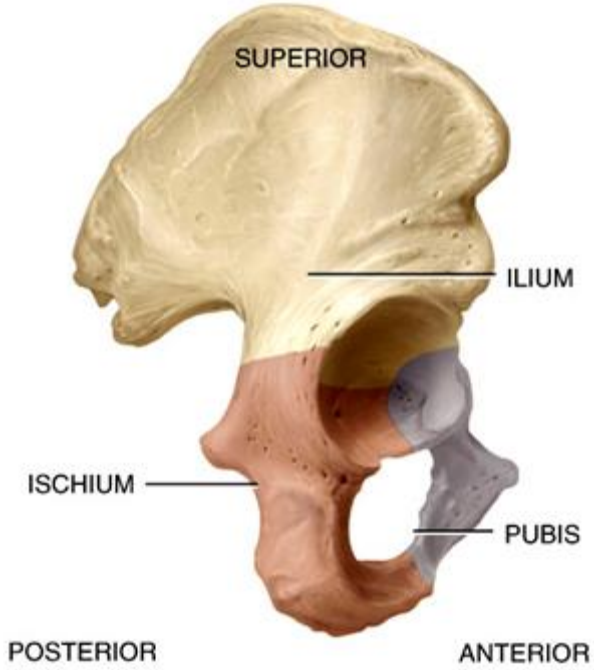
1. **A single *pelvic girdle* (2 bones)**
2. **The *free part (30 bones)***

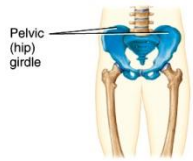


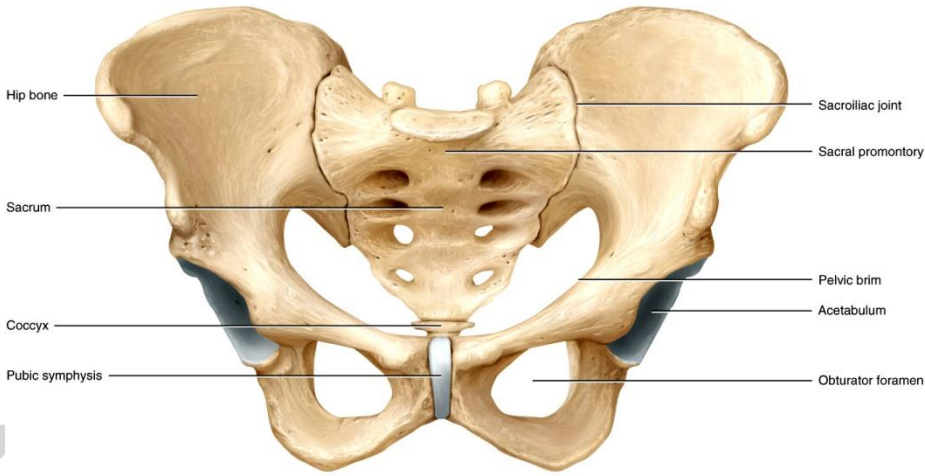
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Pelvic (Hip) Girdle

- Each coxal (hip) bone consists of three bones that fuse together: ilium, pubis, and ischium
- The two coxal bones are joined anteriorly by the pubic symphysis (fibrocartilage)
- Joined posteriorly by the sacrum forming the sacroiliac joints





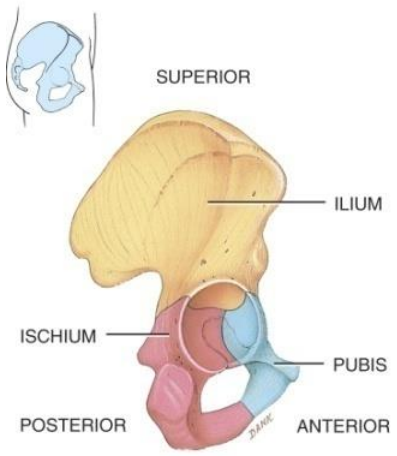


Anterosuperior view of pelvic girdle

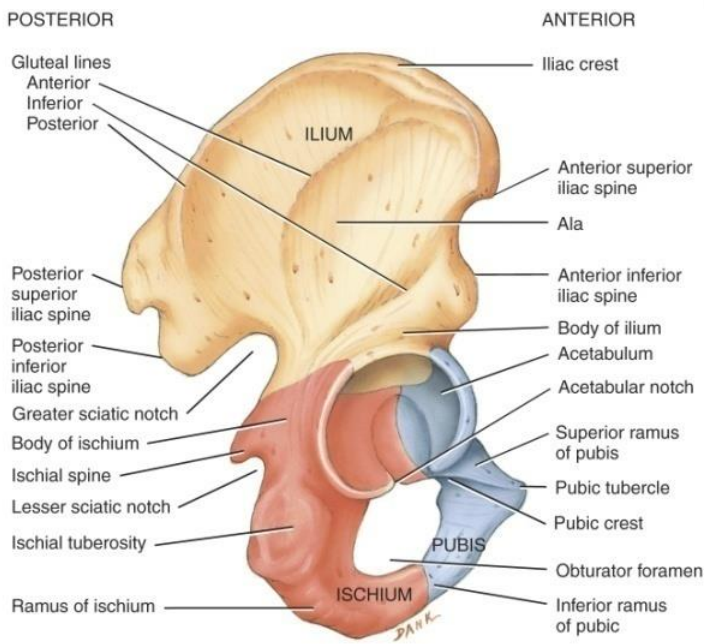
Figure 08.09 Tortora - PAP 12/e
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The Ilium

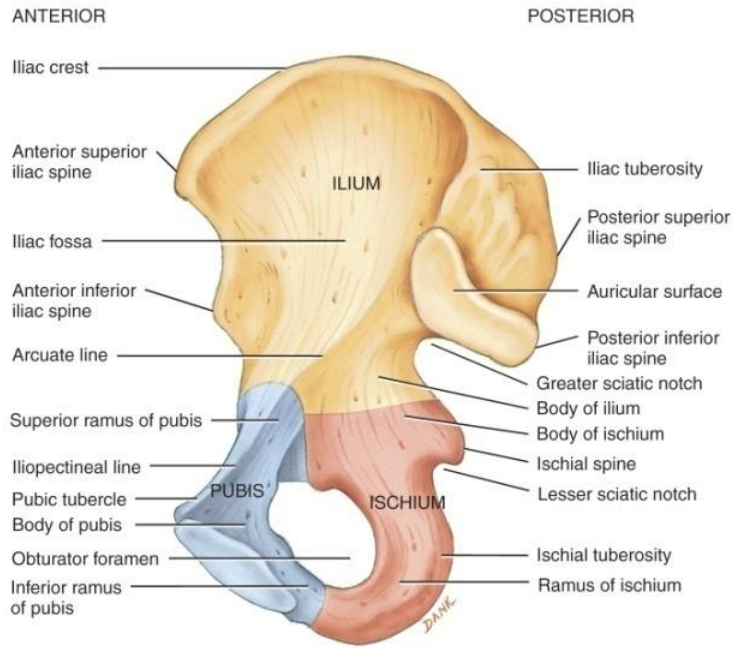
- Largest of the three hip bones
- Ilium is the superior part of the hip bone
- Consists of a superior *ala* and inferior *body* which forms the acetabulum (the socket for the head of the femur)
- Superior border - iliac crest
- *Hip pointer* - occurs at anterior superior iliac spine
- iliac spines
- Greater sciatic notch - allows passage of sciatic nerve
- Iliac fossa for muscle attachment
- Gluteal lines indicating muscle attachment
- Sacroiliac joint at auricular surface & iliac tuberosity



(a) Lateral view showing parts of hip bone



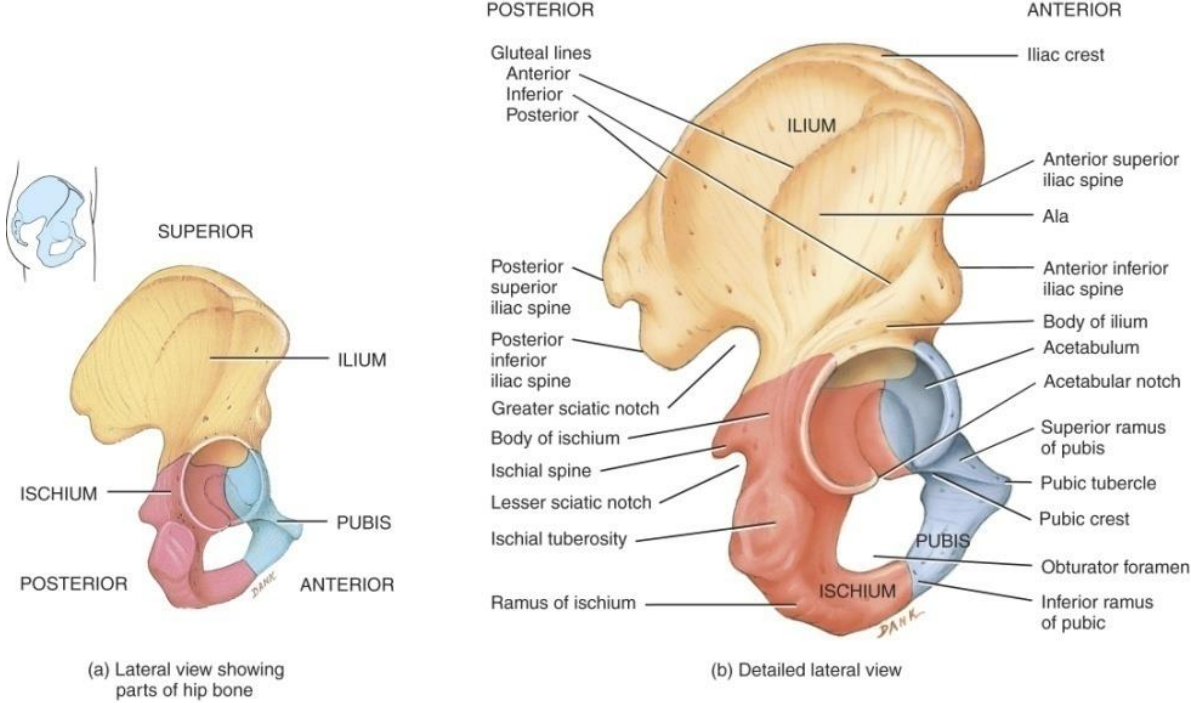
(b) Detailed lateral view



(c) Detailed medial view

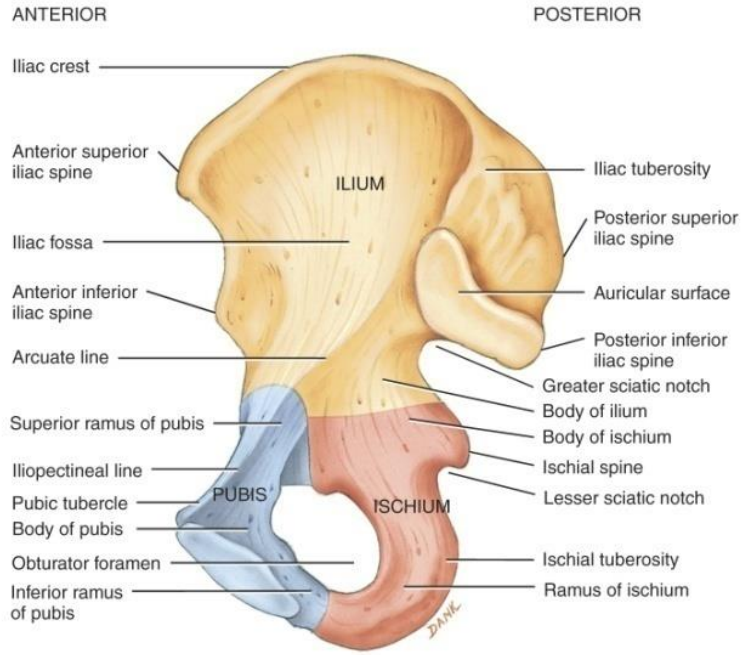
Ischium and Pubis

- Ischium - inferior and posterior part of the hip bone
- Most prominent feature is the ischial tuberosity, it is the part that meets the chair when you are sitting
- ischial spine
- lesser sciatic notch
- ramus
- Pubis - inferior and anterior part of the hip bone
- Superior and inferior rami and body

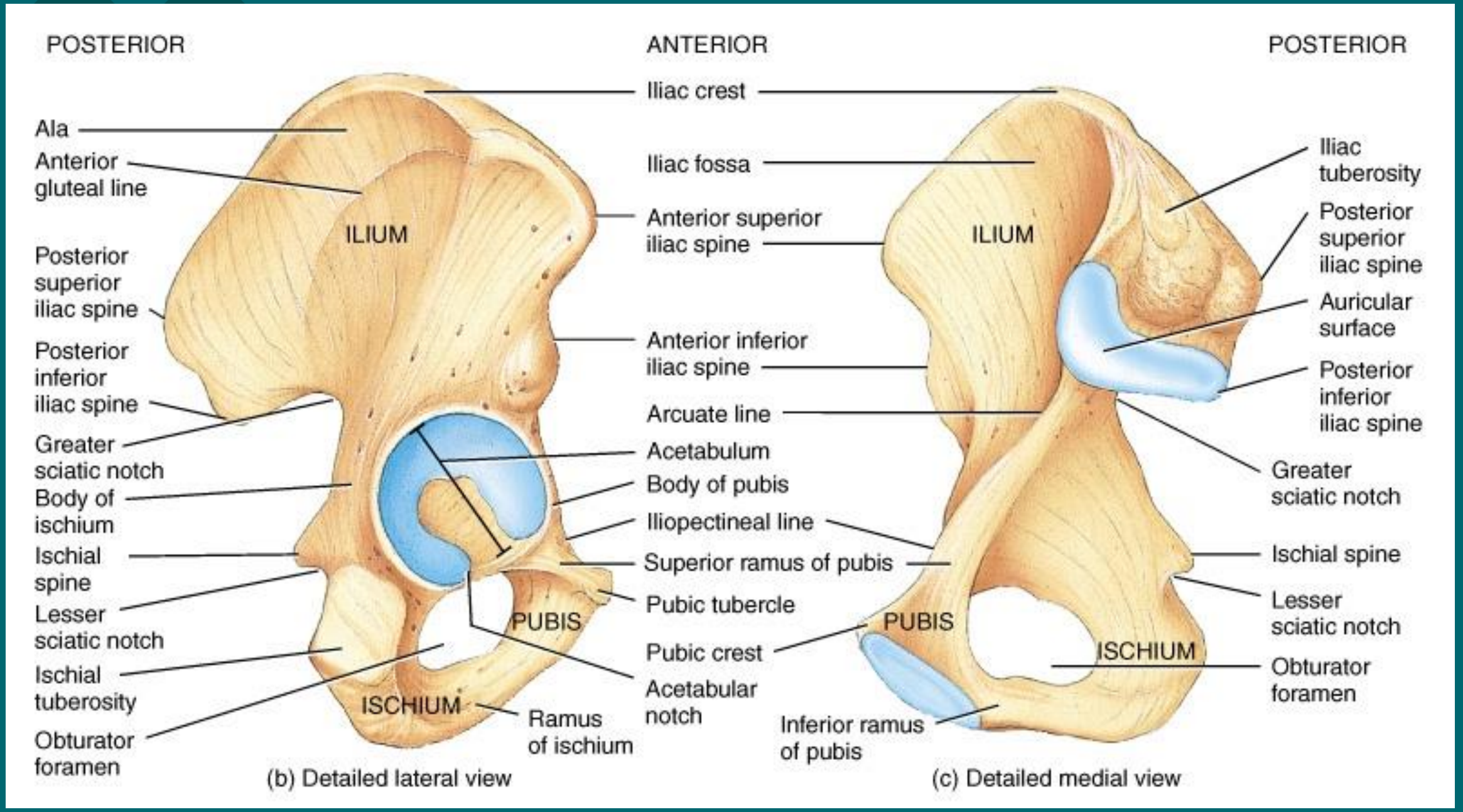


(a) Lateral view showing parts of hip bone

(b) Detailed lateral view

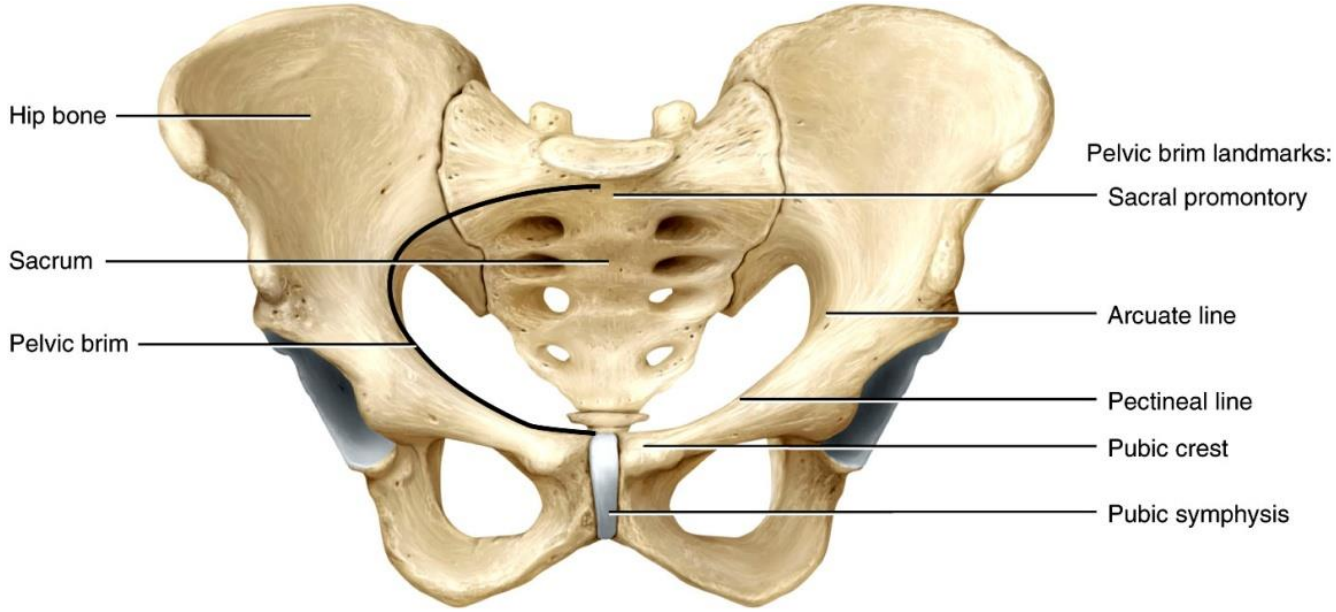


(c) Detailed medial view


POSTERIOR
ANTERIOR
POSTERIOR
Iliac crest
Ala
Anterior gluteal line
Iliac fossa
Anterior superior iliac spine
ILIUM
ILIUM
Iliac tuberosity
Posterior superior iliac spine
Auricular surface
Posterior superior iliac spine
Anterior inferior iliac spine
Posterior inferior iliac spine
Arcuate line
Greater sciatic notch
Acetabulum
Body of pubis
Body of ischium
Iliopectineal line
Ischial spine
Superior ramus of pubis
Lesser sciatic notch
Pubic tubercle
PUBIS
PUBIS
Ischial tuberosity
Pubic crest
ISCHIUM
Obturator foramen
Acetabular notch
ISCHIUM
Ramus of ischium
Obturator foramen
Inferior ramus of pubis
(b) Detailed lateral view
(c) Detailed medial view

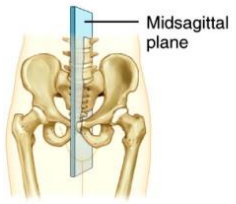
False and True Pelves

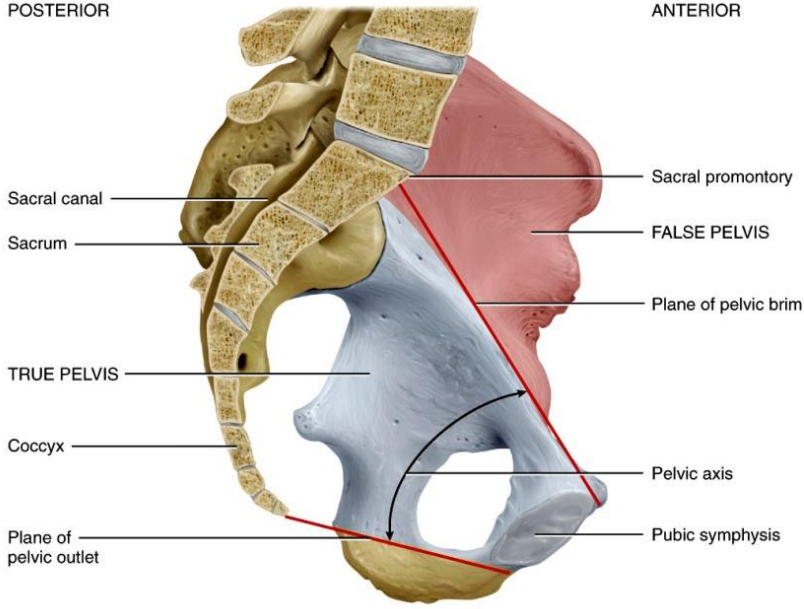
- Pelvic brim - a line from the sacral promontory to the upper part of the pubic symphysis
- False pelvis - lies above this line
- Contains no pelvic organs except urinary bladder (when full) and uterus during pregnancy
- True pelvis - the bony pelvis inferior to the pelvic brim, has an inlet, an outlet and a cavity
- Pelvic axis - path of baby during birth



(a) Anterosuperior view of pelvic girdle

Figure 08.11a Tortora - PAP 12/e
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(b) Midsagittal section indicating locations of true and false pelves

Figure 08.11b Tortora - PAP 12/e
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Comparing Male and Female Pelves

Males •

Bone are larger and heavier •

Pelvic inlet is smaller and heart shaped •

Pubic arch is less the 90° •

Female •

Wider and shallower •

Pubic arch is greater than 90° •

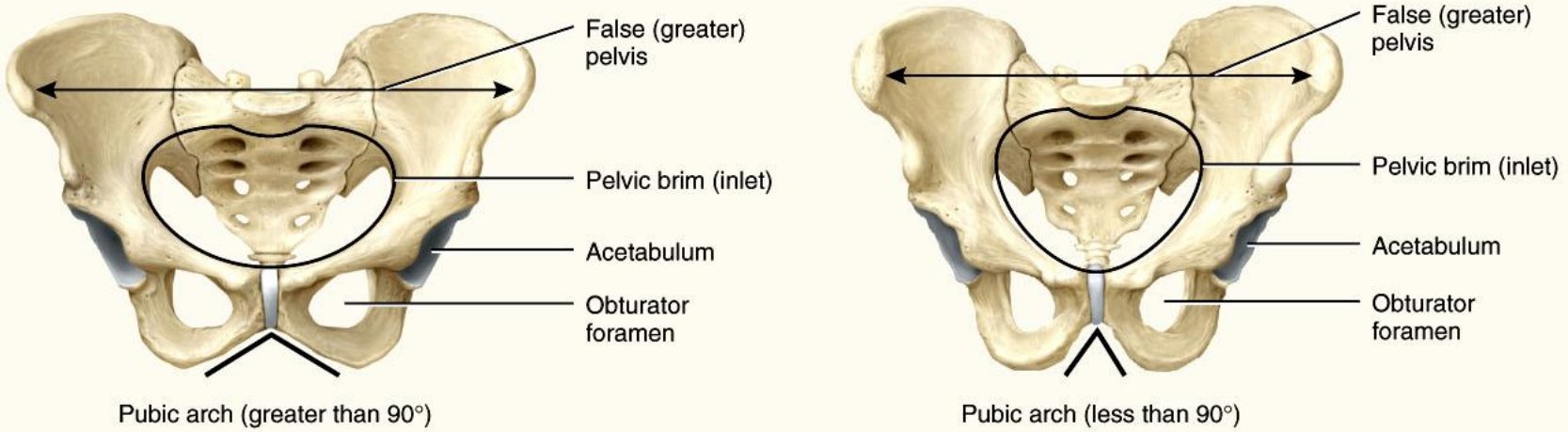
More space in the true pelvis •

Comparing Male and Female Pelves

TABLE 8.1

Comparison of Female and Male Pelves

POINT OF COMPARISON	FEMALE	MALE
General structure	Light and thin.	Heavy and thick.
False (greater) pelvis	Shallow.	Deep.
Pelvic brim (inlet)	Larger and more oval.	Smaller and heart-shaped.
Acetabulum	Small and faces anteriorly.	Large and faces laterally.
Obturator foramen	Oval.	Round.
Pubic arch	Greater than 90° angle.	Less than 90° angle.



Anterior views

TABLE 8.1 CONTINUES

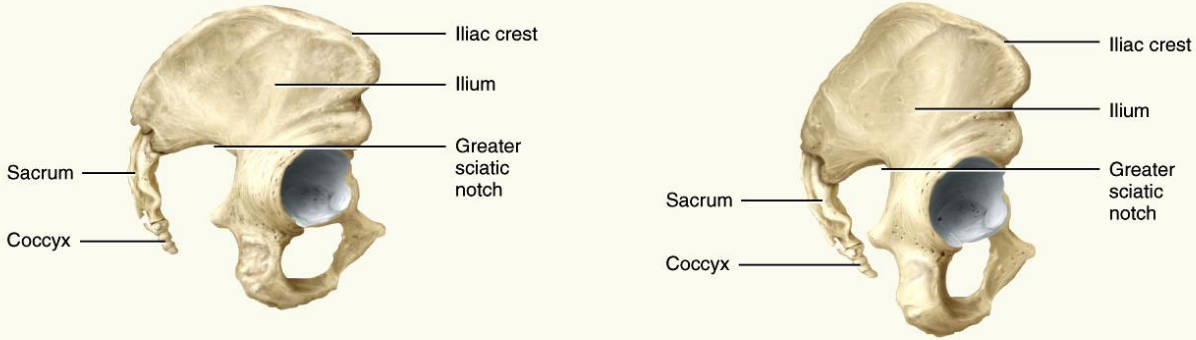
Table 08.01a Tortora - PAP 12/e

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Comparing Male and Female Pelves

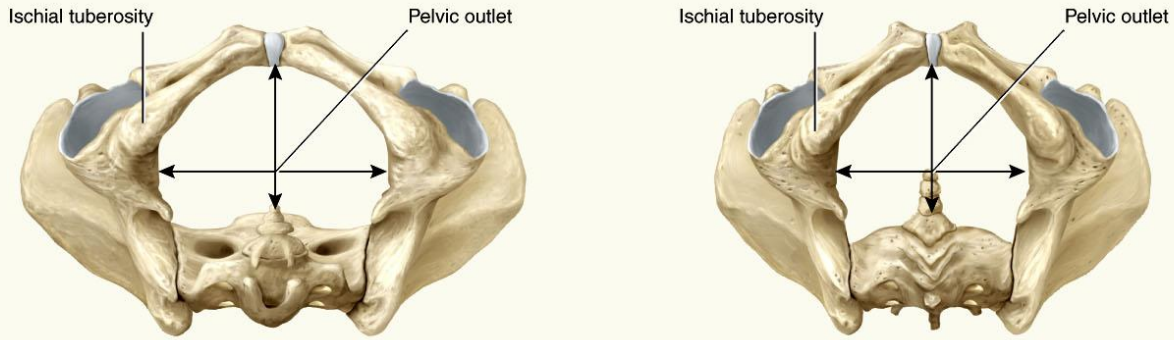
TABLE 8.1 CONTINUED

POINT OF COMPARISON	FEMALE	MALE
Iliac crest	Less curved.	More curved.
Ilium	Less vertical.	More vertical.
Greater sciatic notch	Wide.	Narrow.
Coccyx	More movable and more curved anteriorly.	Less movable and less curved anteriorly.
Sacrum	Shorter, wider (see anterior views), and less curved anteriorly.	Longer, narrower (see anterior views), and more curved anteriorly.



Right lateral views

Pelvic outlet	Wider.	Narrower.
Ischial tuberosity	Shorter, farther apart, and more medially projecting.	Longer, closer together, and more laterally projecting.



Inferior views

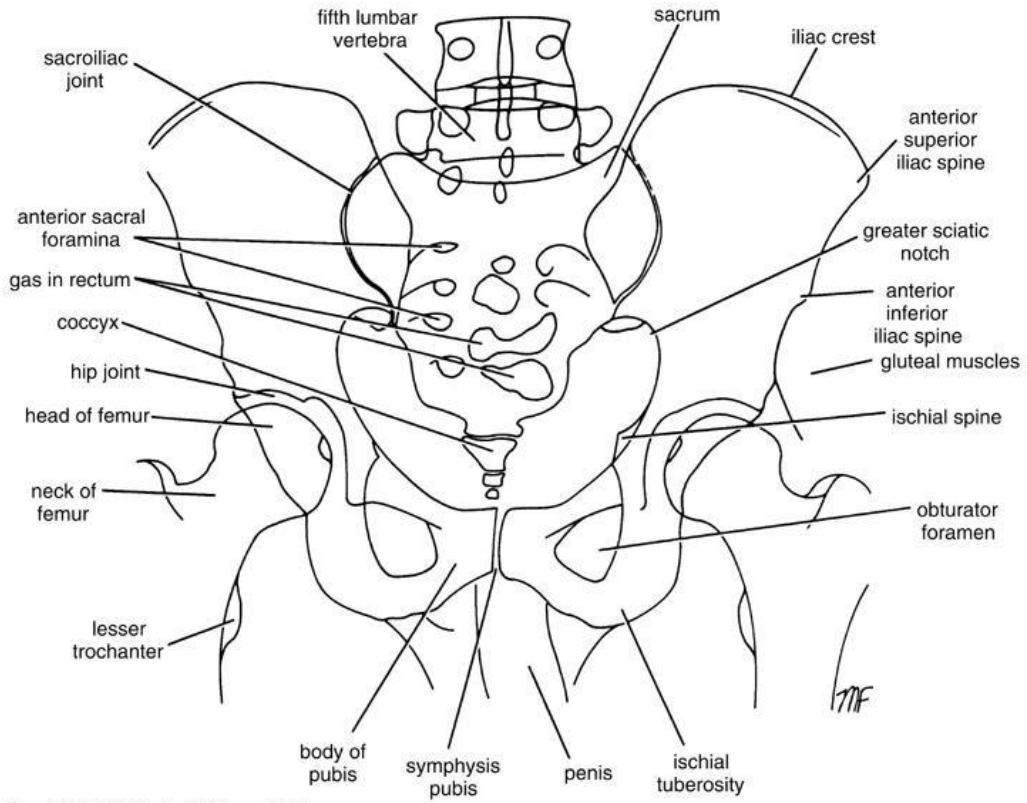
Table 08.01b Tortora - PAP 12/e

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Male pelvis



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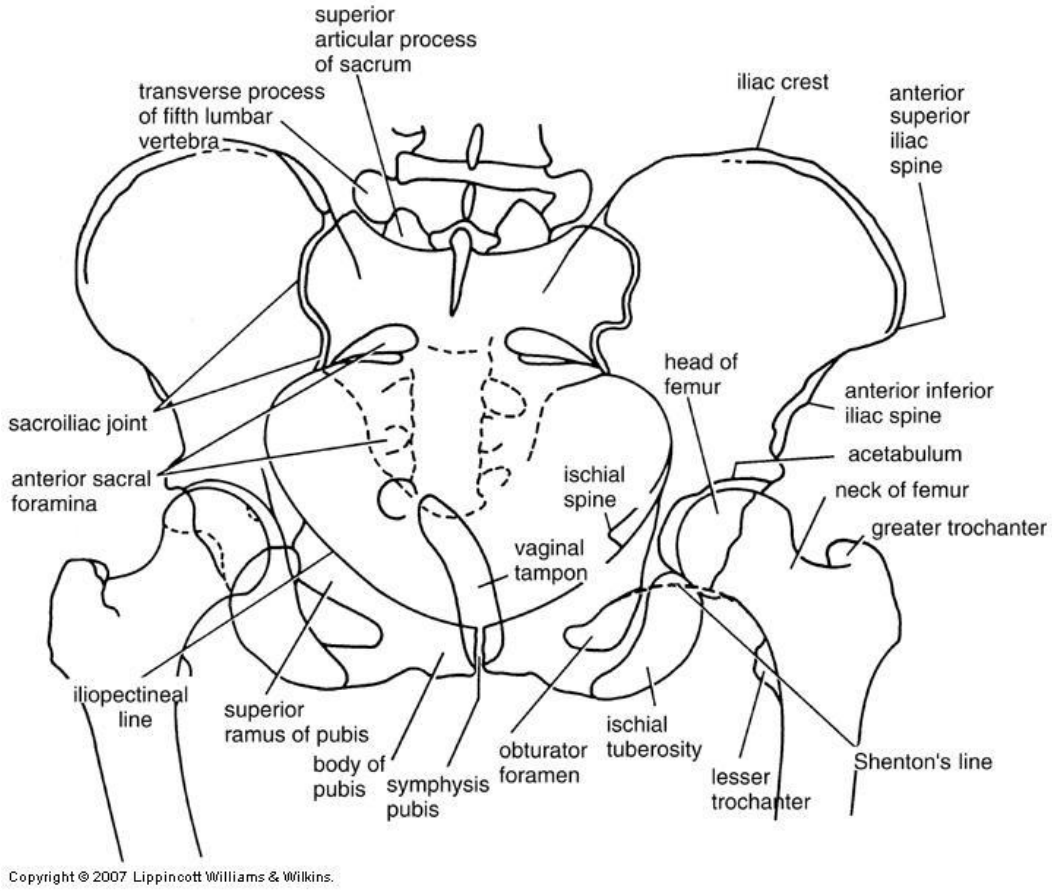


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Female pelvis



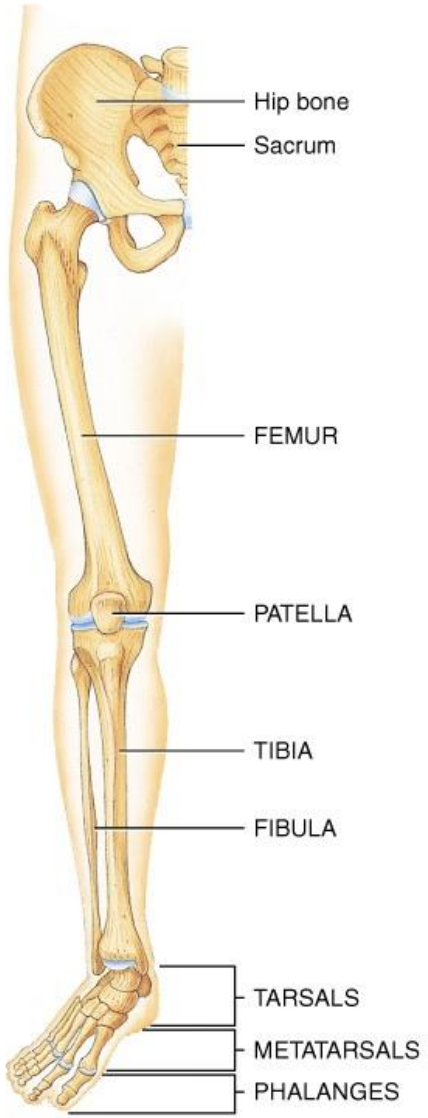
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Lower Extremity

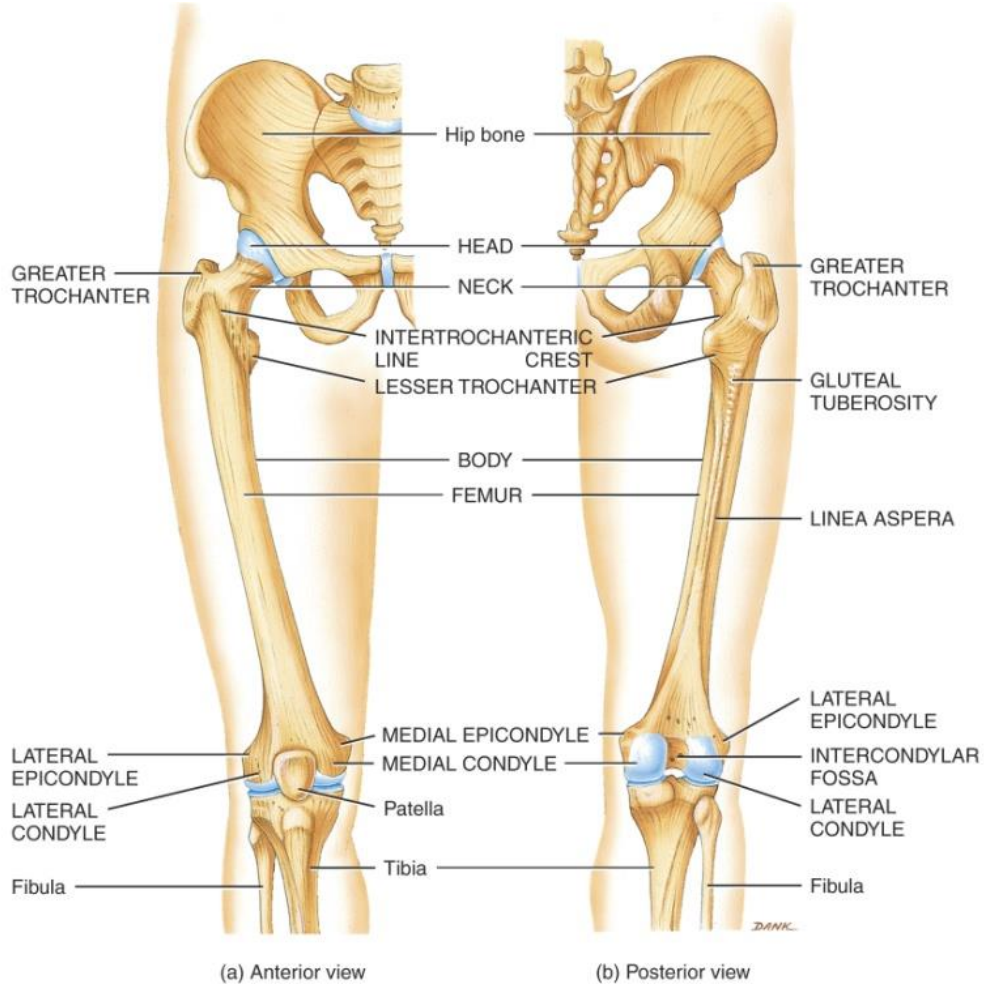
- Each lower limb = 30 bones
  - femur and patella within the thigh
  - tibia & fibula within the leg
  - tarsal bones in the foot
  - metatarsals within the forefoot
  - phalanges in the toes
- Joints
  - hip, knee, ankle
  - proximal & distal tibiofibular
  - metatarsophalangeal

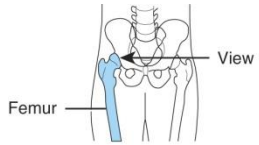


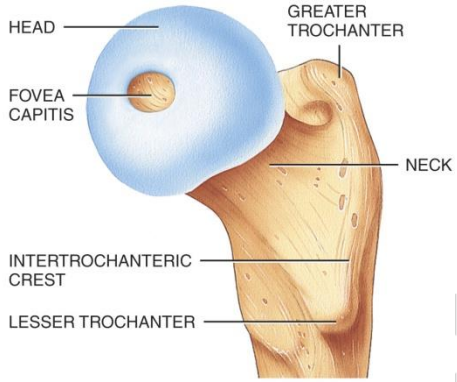
Femur (thighbone)

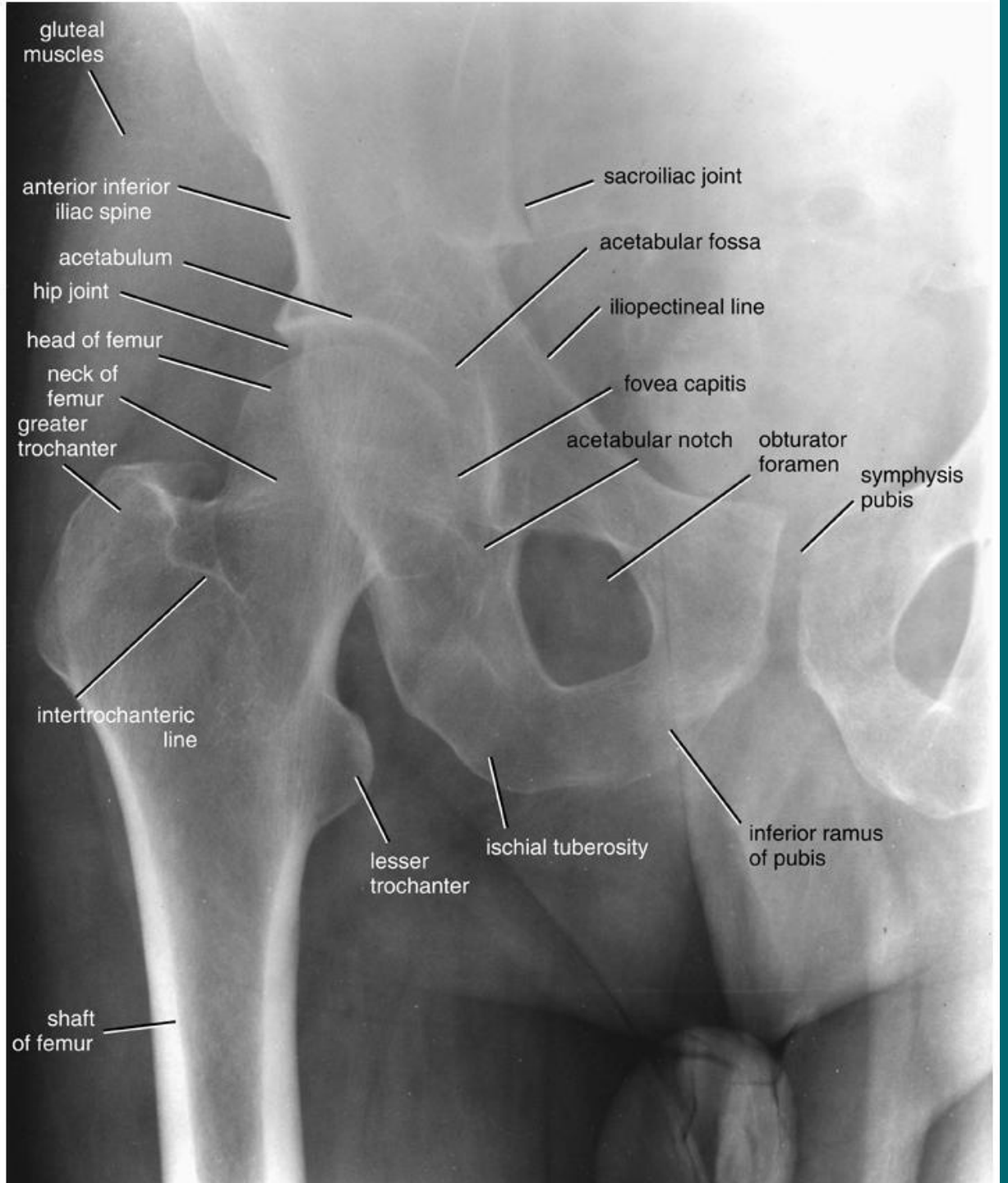
- longest & strongest bone in body
- head articulates with acetabulum (attached by ligament of head of femur)
  - Fovea capitis
- neck is common fracture site
- greater & lesser trochanters, linea aspera, & gluteal tuberosity-- muscle attachments
- medial & lateral condyles articulate with tibia
- patellar surface anteriorly between condyles
- intercondylar fossa - depression between the condyles










gluteal muscles
anterior inferior iliac spine
acetabulum
hip joint
head of femur
neck of femur
greater trochanter
intertrochanteric line
shaft of femur
lesser trochanter
sacroiliac joint
acetabular fossa
iliopectineal line
fovea capitis
acetabular notch
obturator foramen
symphysis pubis
ischial tuberosity
inferior ramus of pubis

Patella

- Largest sesamoid bone in the body
- Forms the patellofemoral joint
- Superior surface is the base
- Inferior, narrower surface is the apex
- Thick articular cartilage lines the posterior surface
- Increases the leverage of the quadriceps femoris muscle

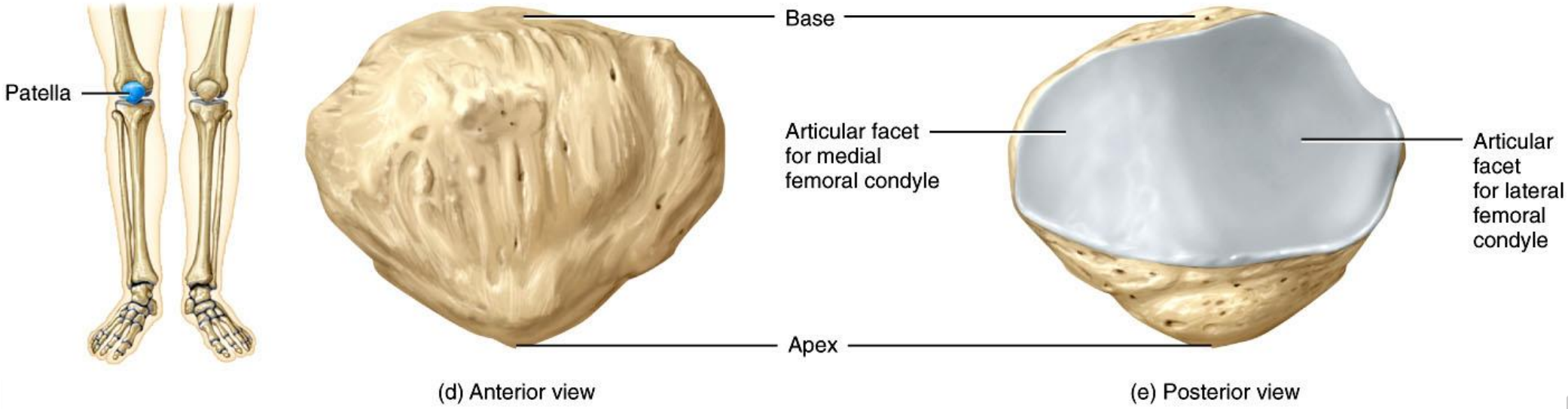


Figure 08.14 Tortora - PAP 12/e

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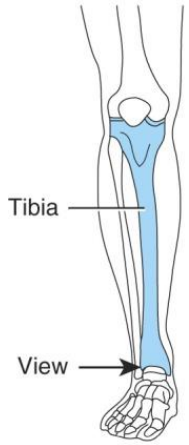
Tibia and Fibula

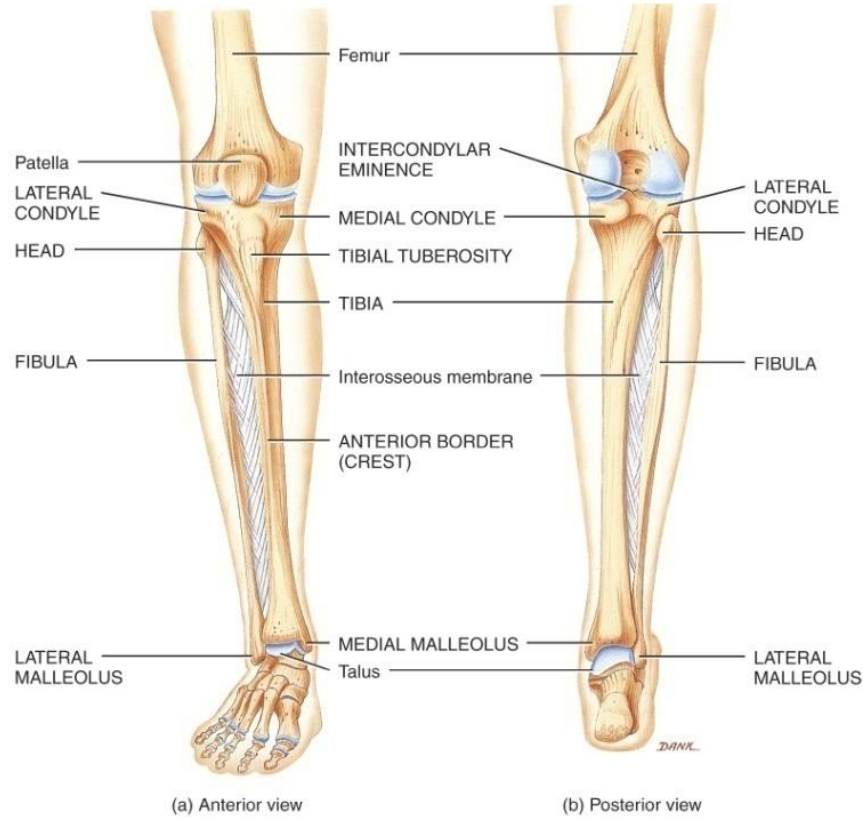
Tibia •
medial & larger bone of leg •
weight-bearing bone •
lateral & medial condyles •
tibial tuberosity for patellar lig. •
proximal tibiofibular joint •
medial malleolus at ankle •
fibular notch •

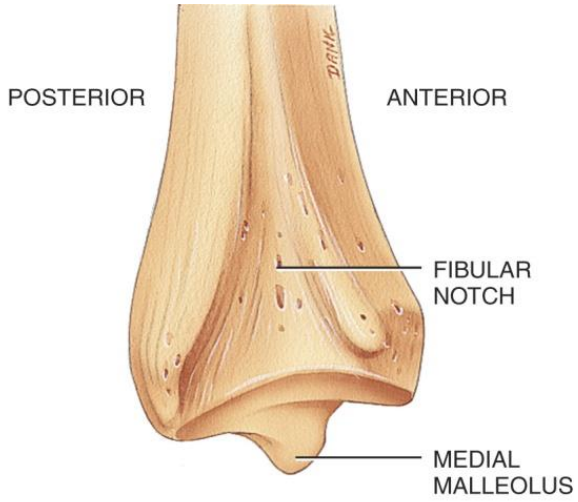
Fibula •
not part of knee joint •
muscle attachment only •
lateral malleolus at ankle •

MNEMONIC for location of tibia and fibula:
The fibu**LA** is **LA**teral.

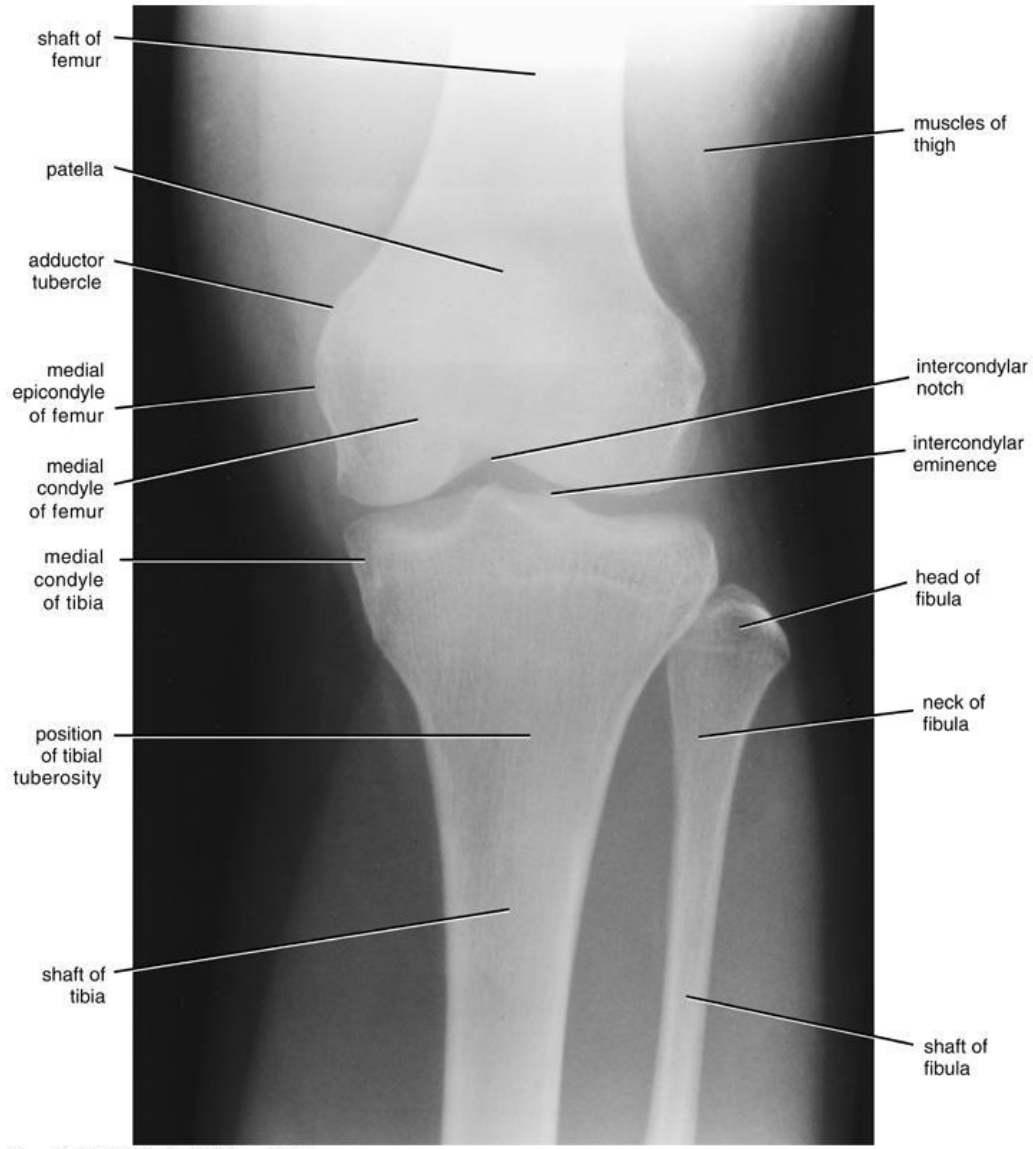




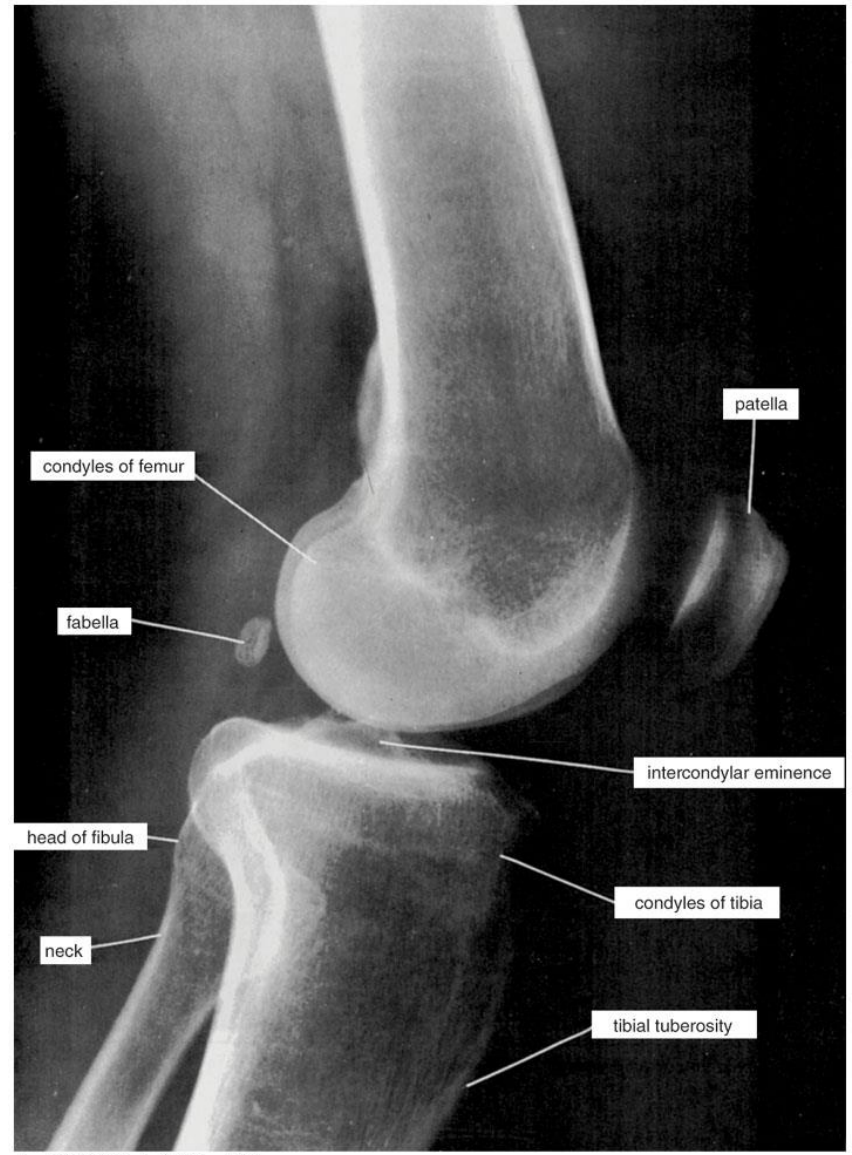




(c) Lateral view of distal end of tibia



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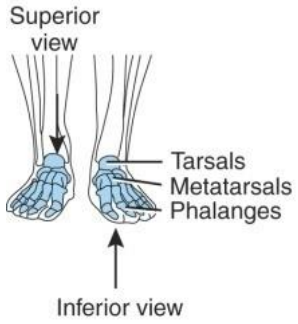


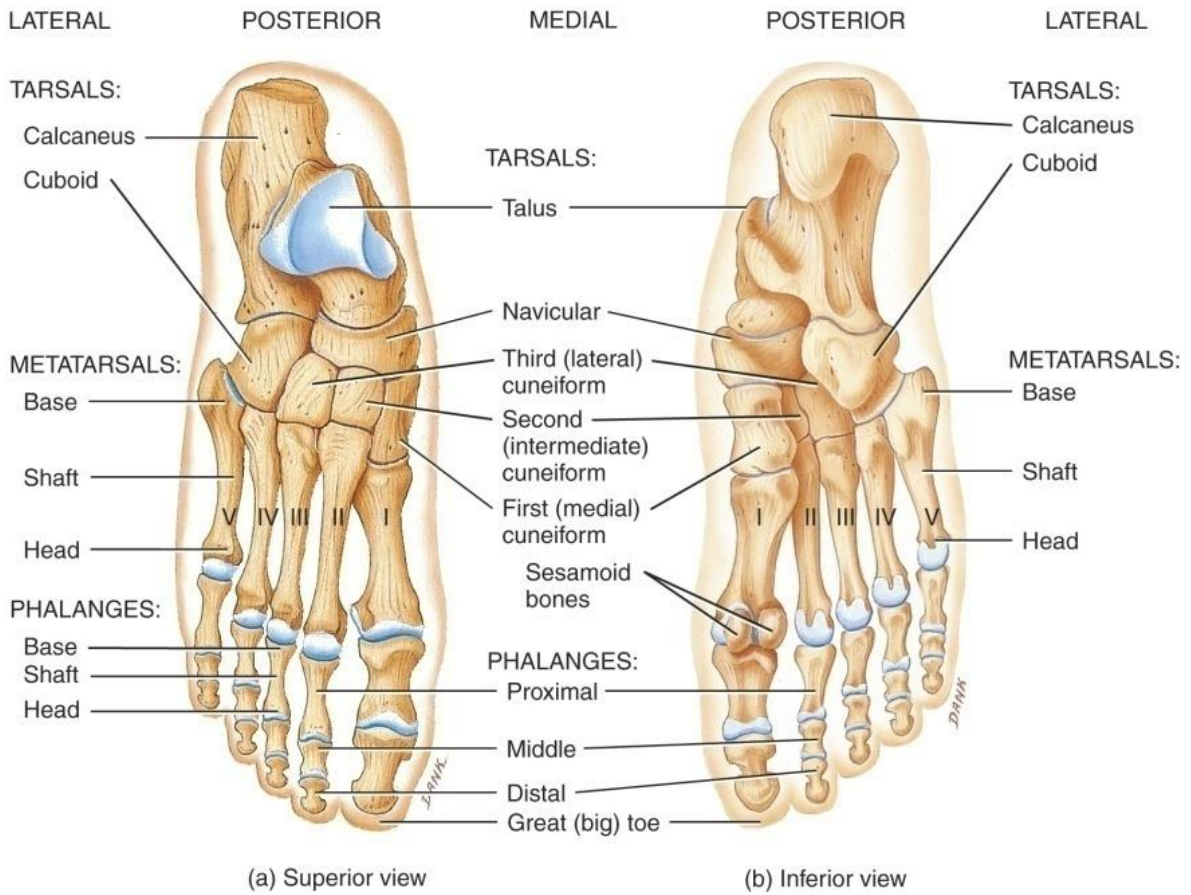
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Tarsus

- Proximal region of foot (contains 7 tarsal bones)
- Talus = ankle bone (articulates with tibia & fibula)
- Calcaneus - heel bone
- Cuboid, navicular & 3 cuneiforms





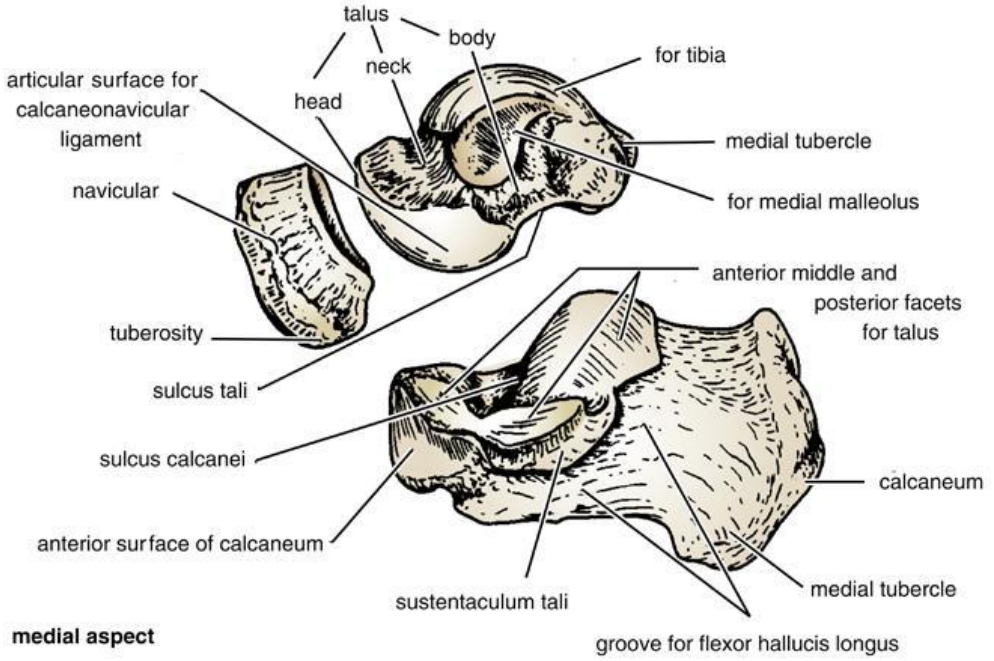
MNEMONIC for tarsal bones:

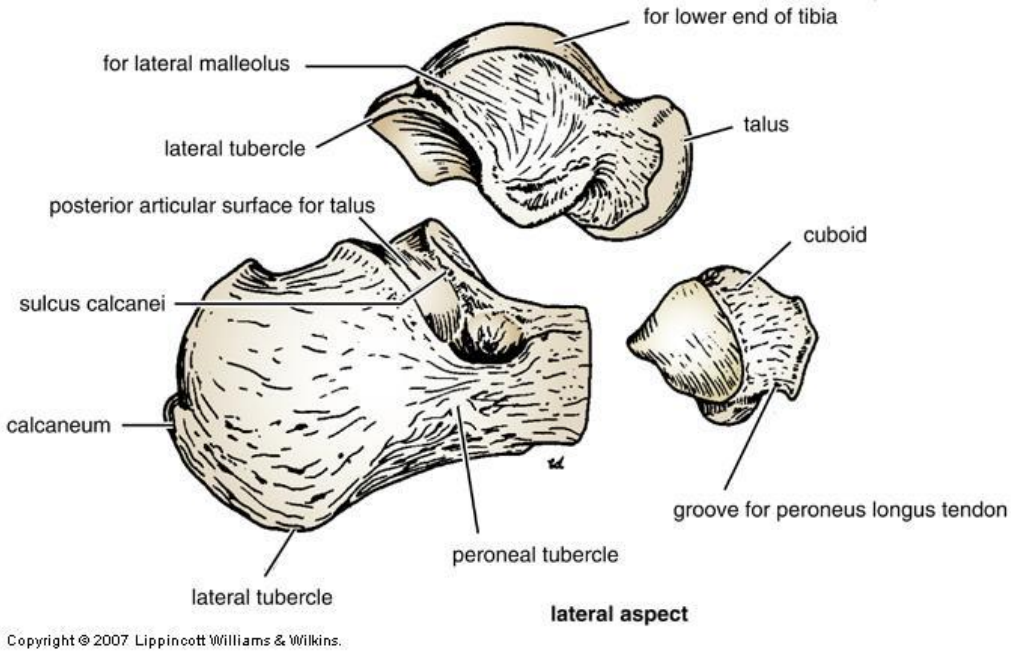
Tall **C**enters **N**ever **T**ake **S**hots **F**rom **C**orners.

Talus **C**alcaneus **N**avicular **T**hird cuneiform **S**econd cuneiform **F**irst cuneiform **C**uboid

Tarsus

- Calcaneum
  - Surfaces
  - Tubercles
  - Sustentaculum tali
  - Sulcus calcanei
  - Peroneal tubercle
- Talus
  - Head, neck, and body
  - Sulcus tali
  - Sinus tarsi
  - Interosseous talocalcaneal ligament

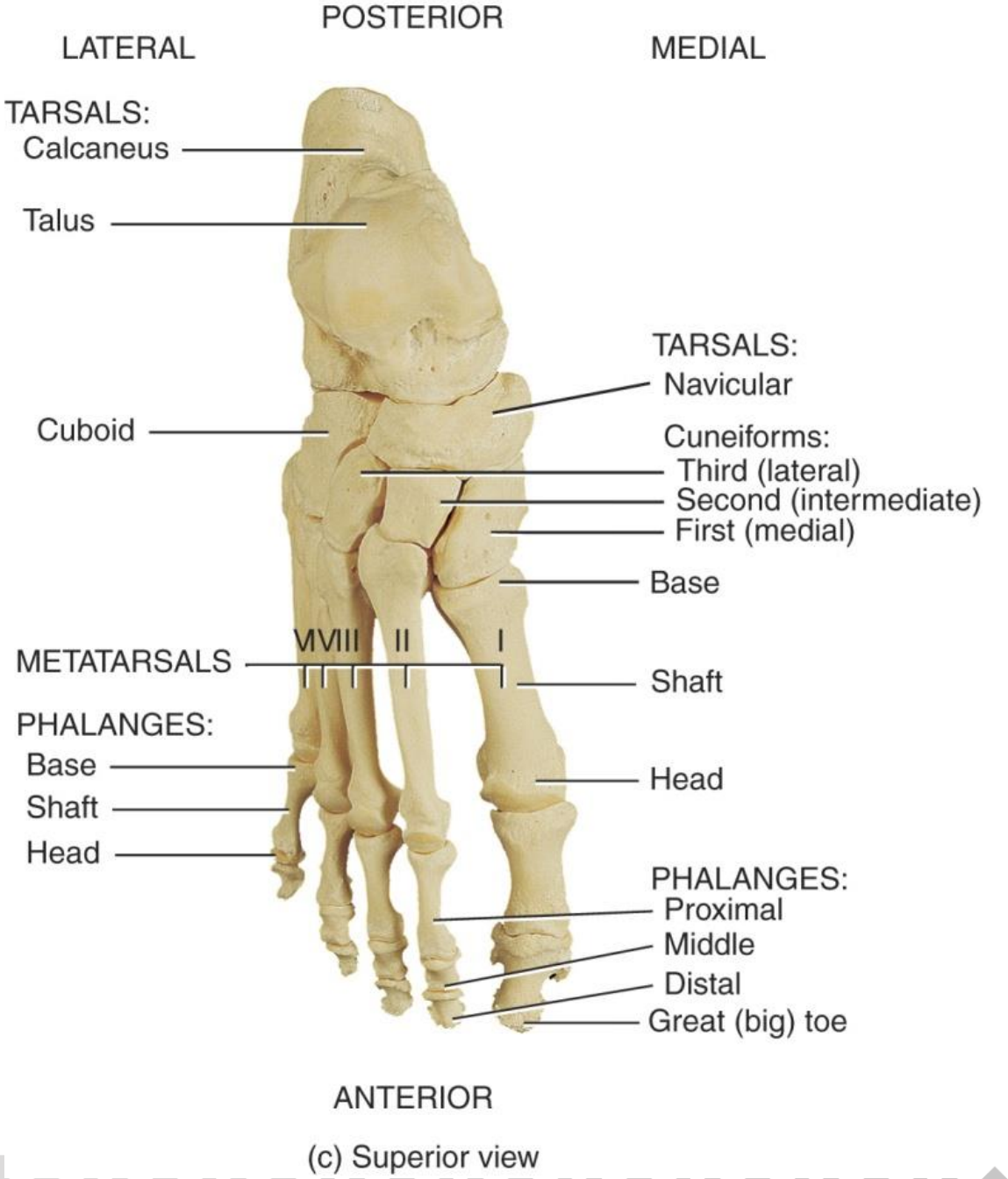




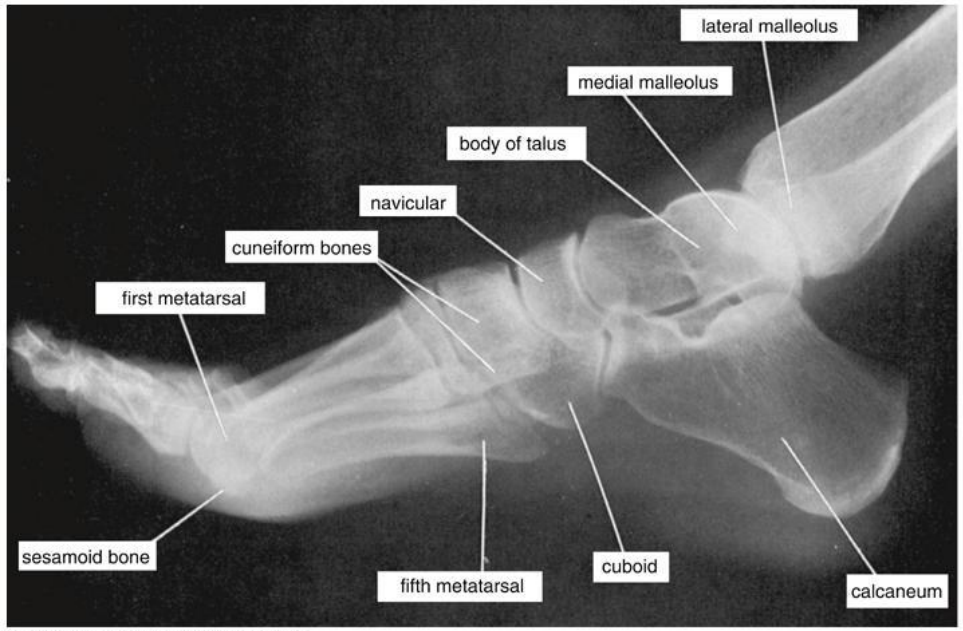
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Metatarsus and Phalanges

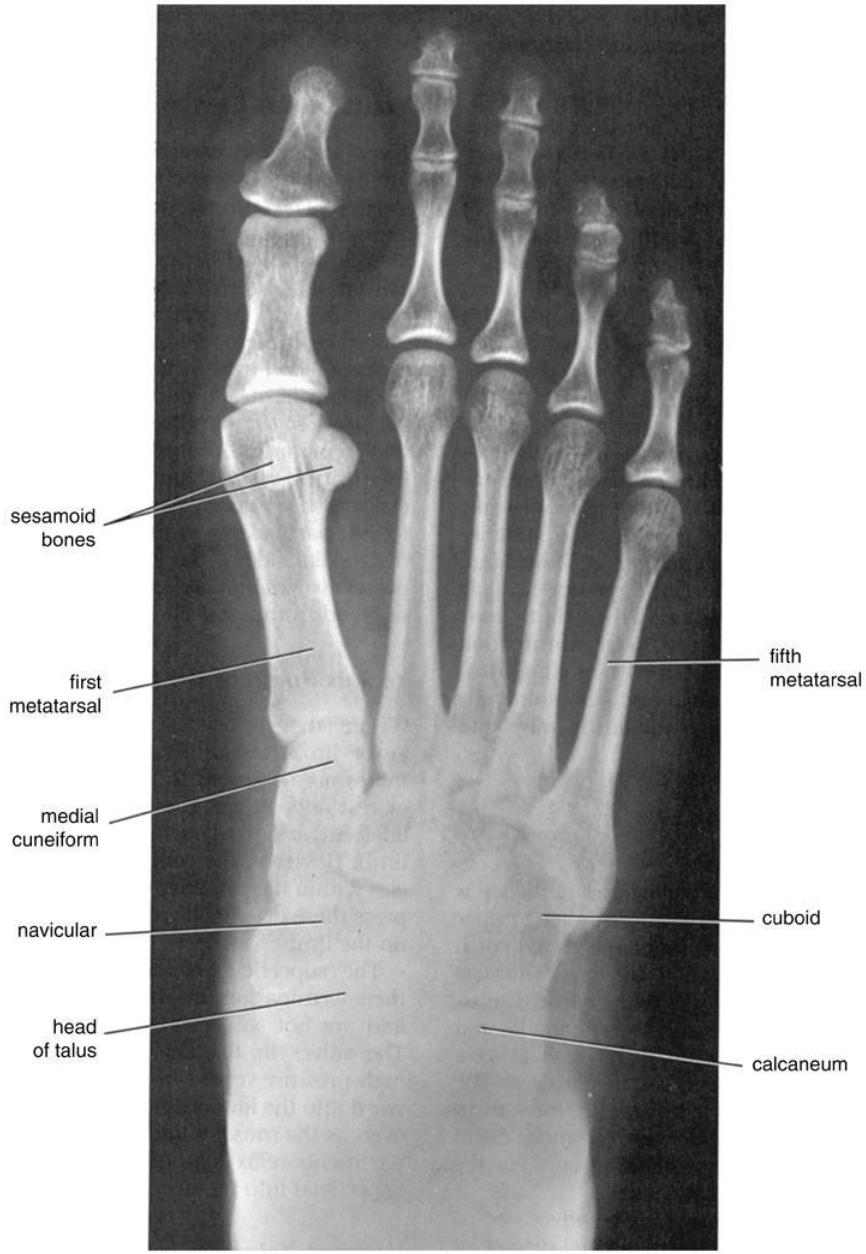
- Metatarsus
  - midregion of the foot
  - 5 metatarsals (1 is most medial)
  - each with base, shaft and head
- Phalanges
  - distal portion of the foot
  - similar in number and arrangement to the hand
  - big toe is hallux



(c) Superior view



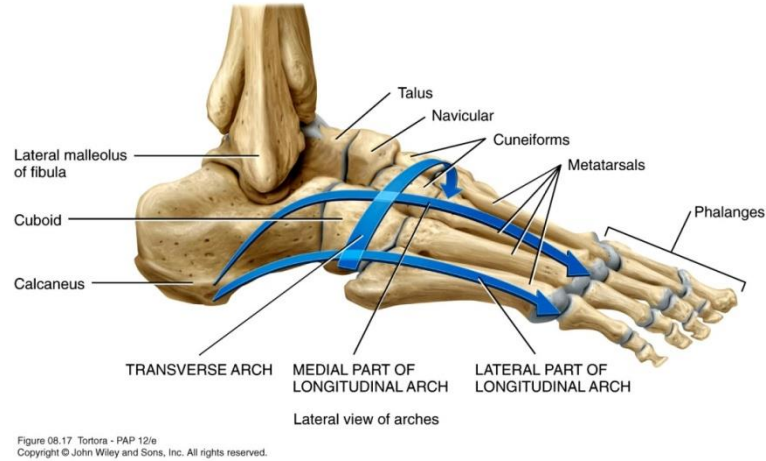
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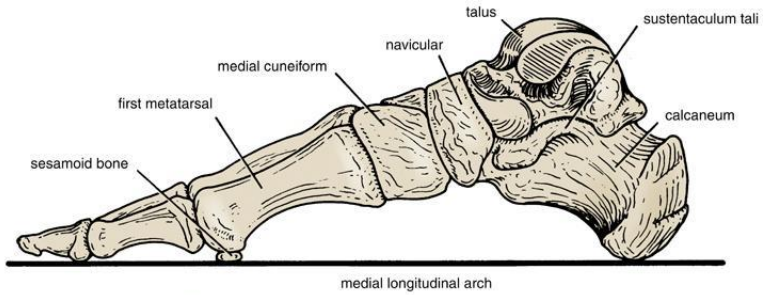
Arches of the Foot

- Function
  - distribute body weight over foot
  - yield & spring back when weight is lifted
- Longitudinal arches along each side of foot
  - Medial longitudinal arch
    - Calcaneum, talus, navicular, cuniforms, and 1-3 metatarsals
  - Lateral longitudinal arch
    - Calcaneum, cuboid, and 4-5 metatarsals
- Transverse arch across midfoot region
  - Cuboid, cuneiforms & bases of metatarsals
- Flatfoot - the arches decrease or "fall"
- Clawfoot - too much arch occurs due to various pathologies

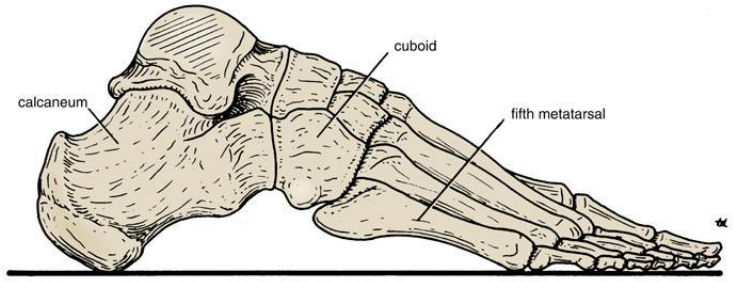


Lateral view of arches

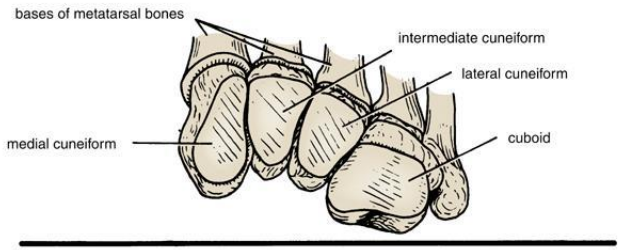
Figure 08.17 Tortora - PAP 12/e
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medial longitudinal arch



lateral longitudinal arch



transverse arch

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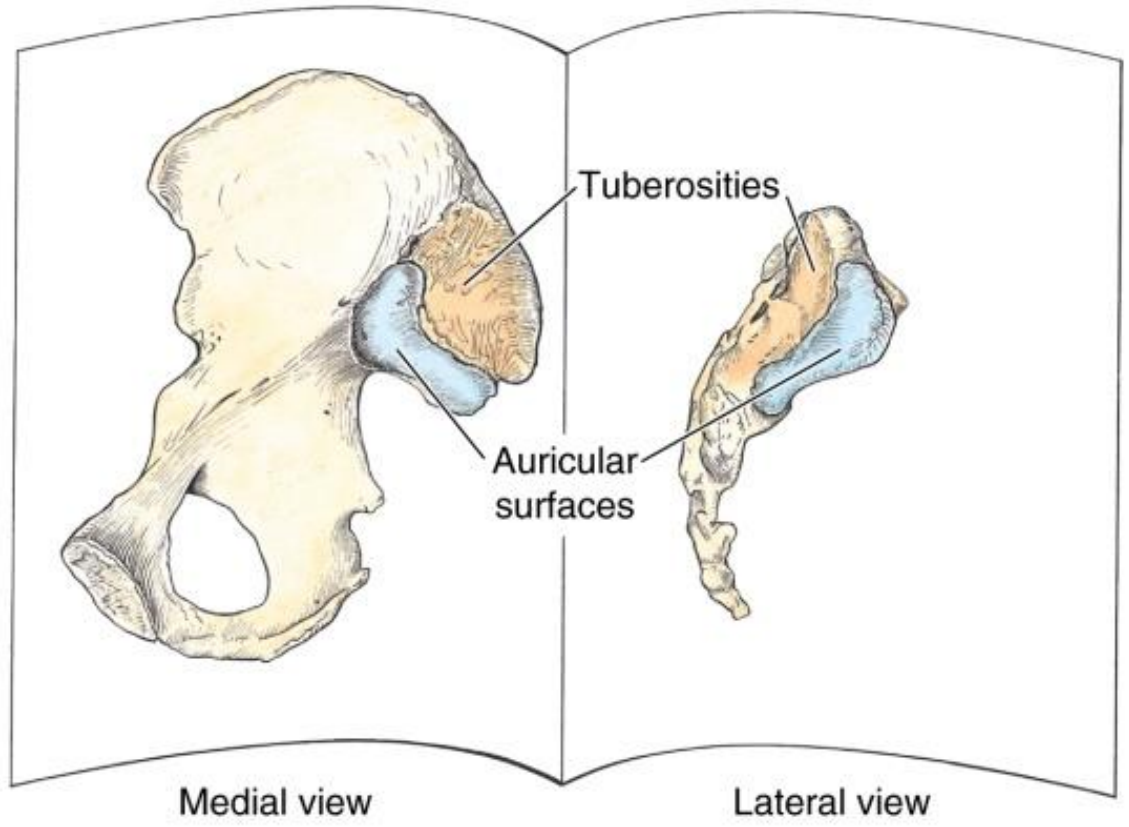


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Lower Limb Joints

Sacroiliac Joint

- Auricular surfaces of the sacrum and the iliac bone
- No movement; transmit body weight from vertebral column to pelvis
- In elderly people synovial cavity disappear and becomes fibrous joint

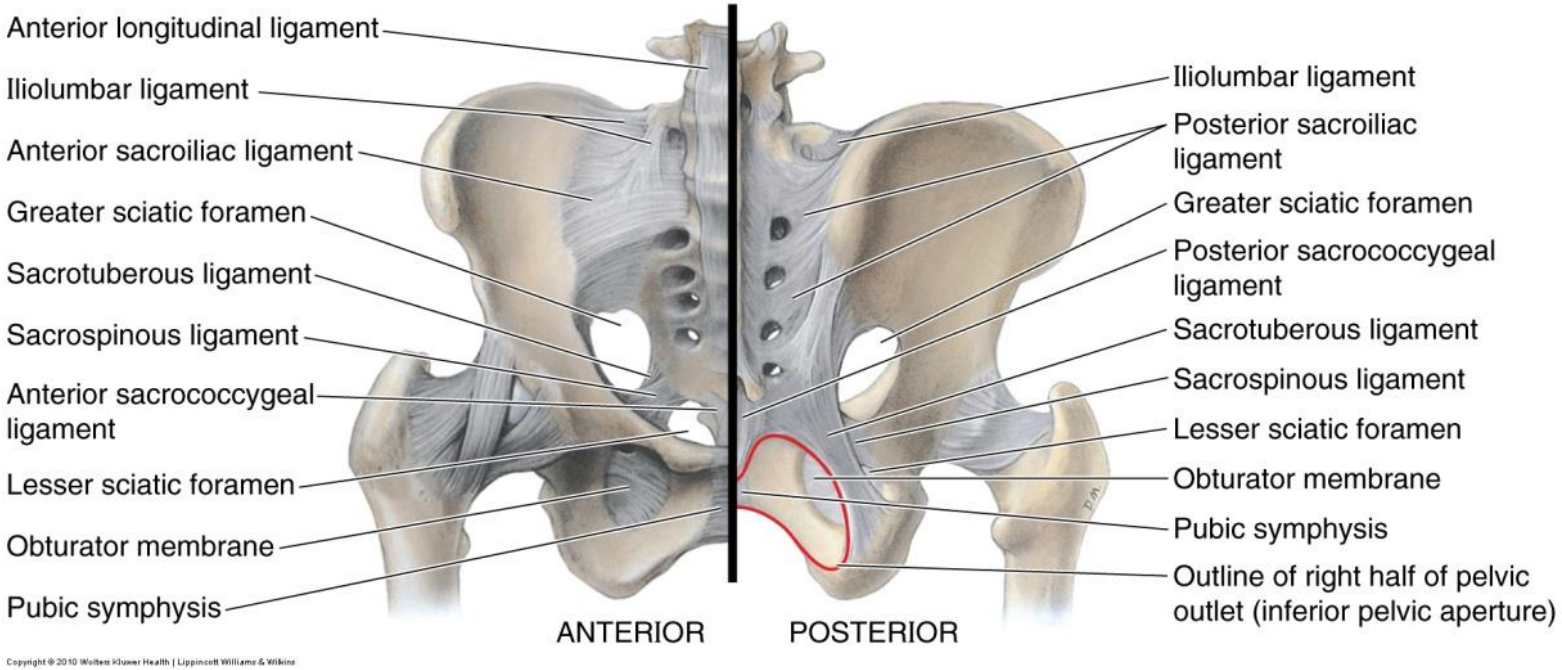


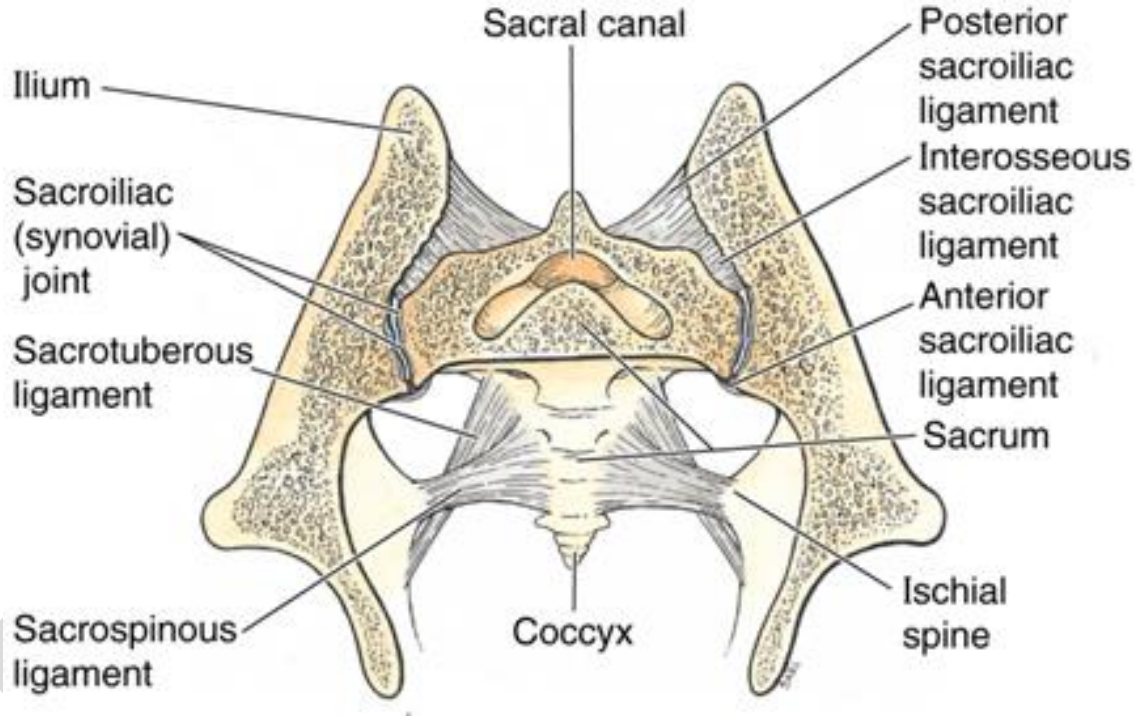
(B) Open book view of articulating surfaces of sacroiliac joint

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Sacroiliac Joint

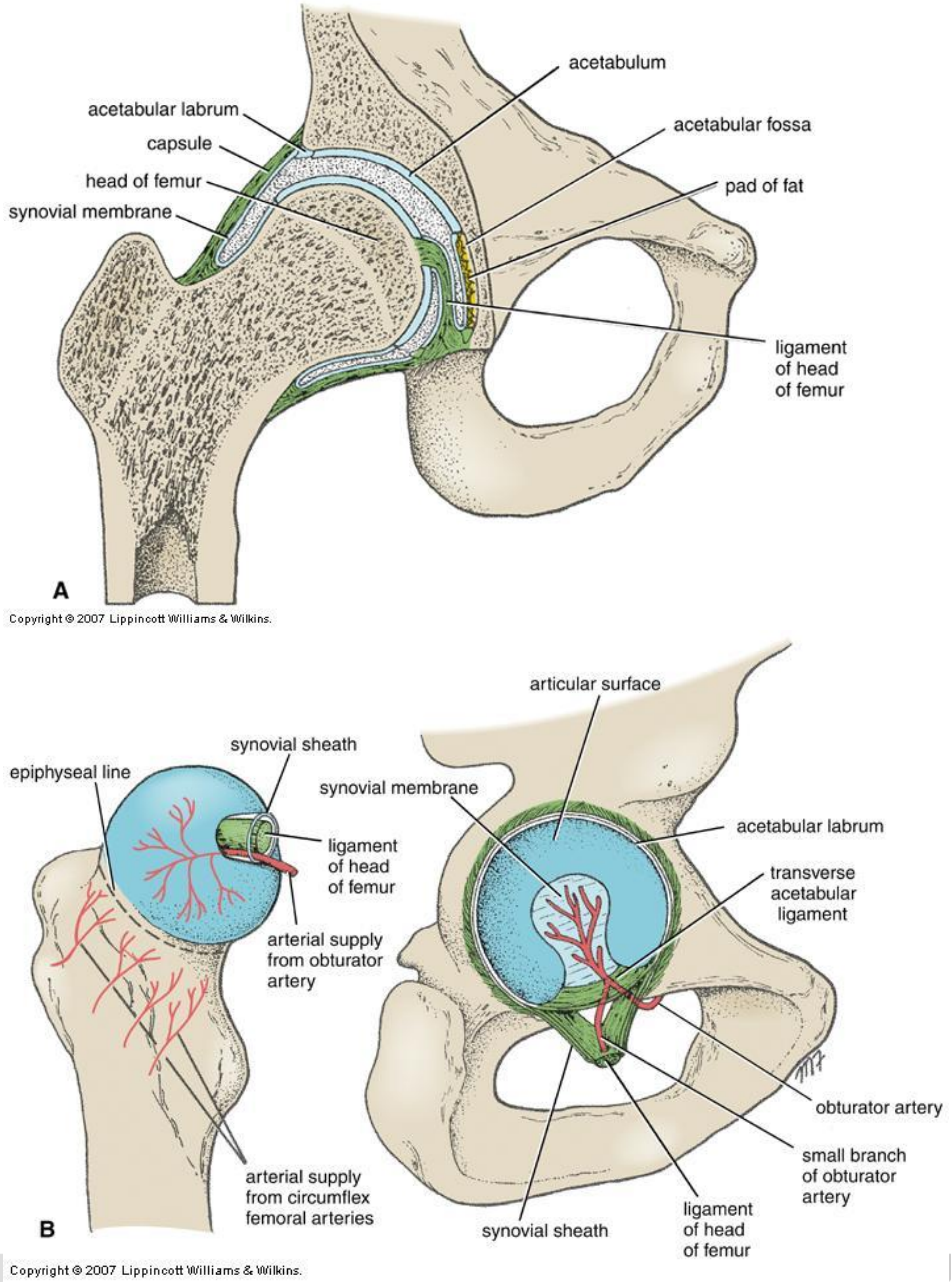
- Associated ligaments
  - Posterior sacroiliac ligament
  - Interosseous sacroiliac ligament (between tuberosities of sacrum and iliac bone)
  - Anterior sacroiliac ligament
- Accessory ligaments
  - Sacrotuberous ligament
    - greater sciatic foramen
  - Sacrospinous ligament
    - lesser sciatic foramen
  - Iliolumbar ligament
- Nerve supply: sacral spinal nerves





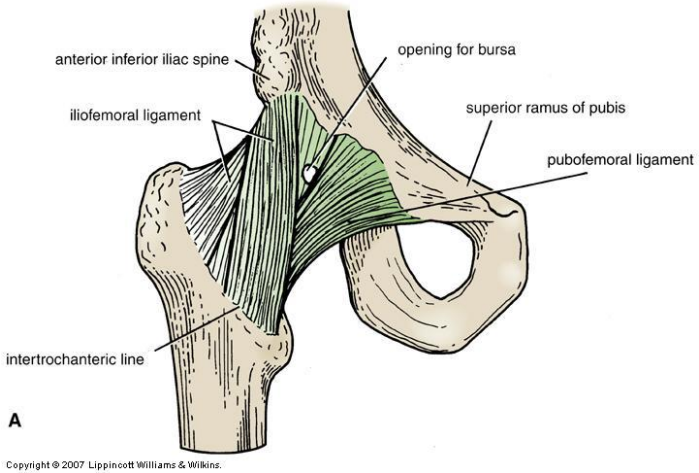
Hip Joint

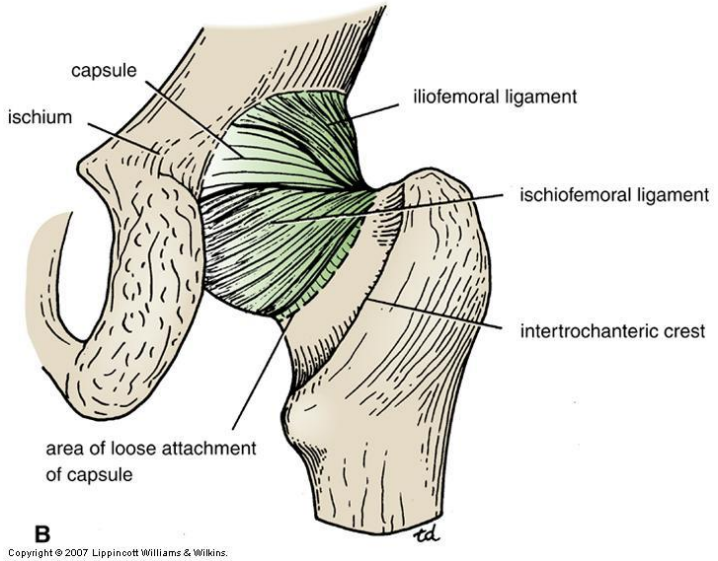
- Head of the femur and the acetabulum
- Structures
  - Acetabular notch
  - Transverse acetabular ligament
  - Acetabular labrum
- Ball and socket joint
- All movements
- Nerve supply: femoral, obturator, and sciatic nerves and nerve to the quadratus femoris m.

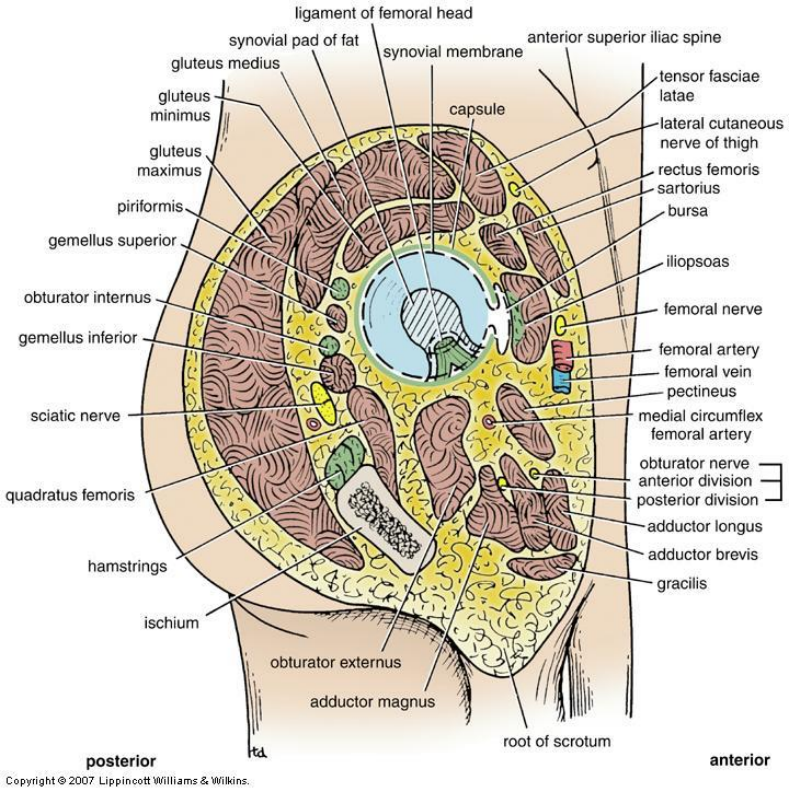


Hip Joint

- Ligaments
  - Extracapsular ligaments
    - Iliofemoral ligament
    - Pubofemoral ligament
    - Ischiofemoral ligament
  - Intracapsular ligament
    - Transverse acetabular ligament
    - Ligament of the head of the femur
- Psoas bursa
- Relations

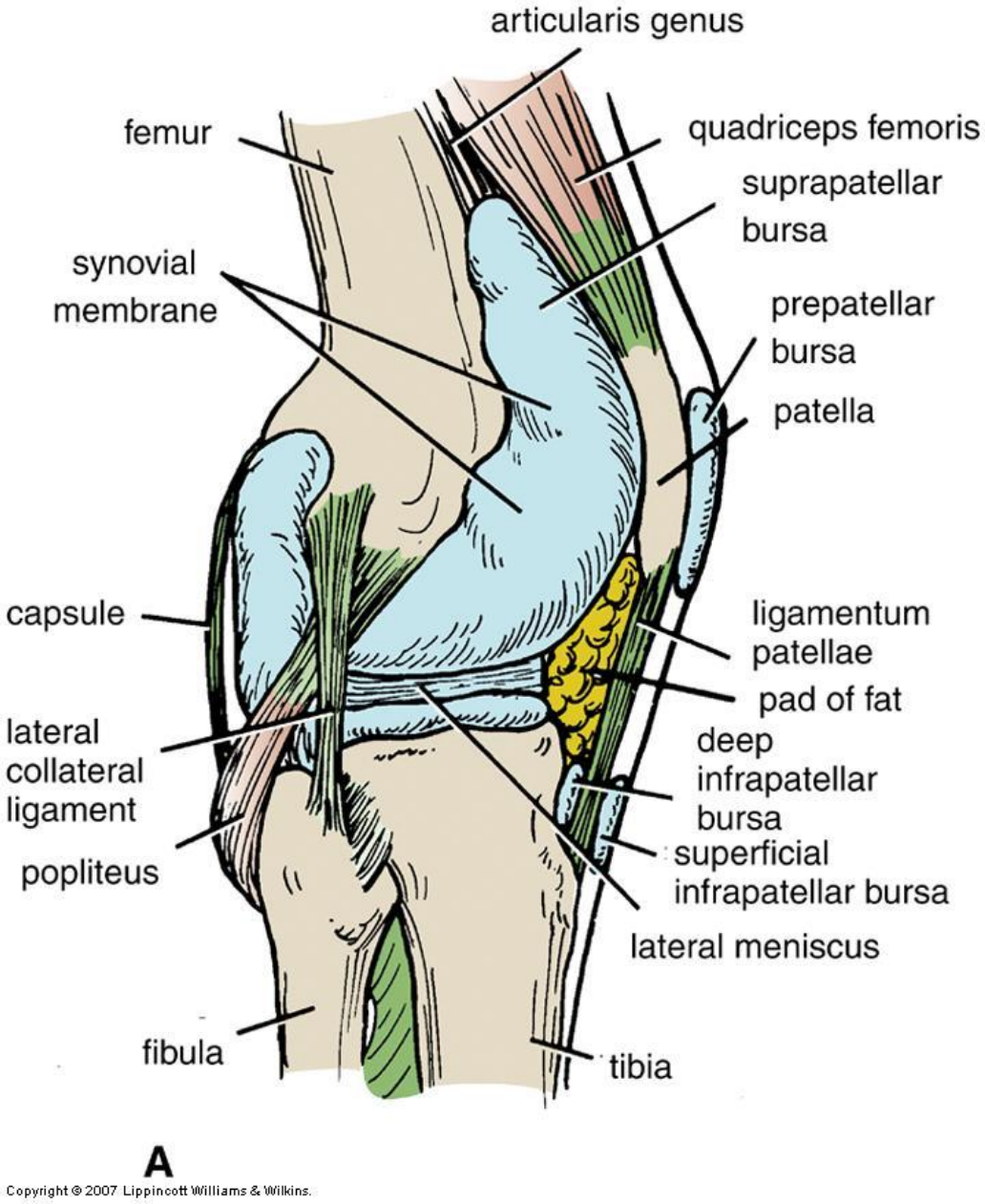






Knee Joint

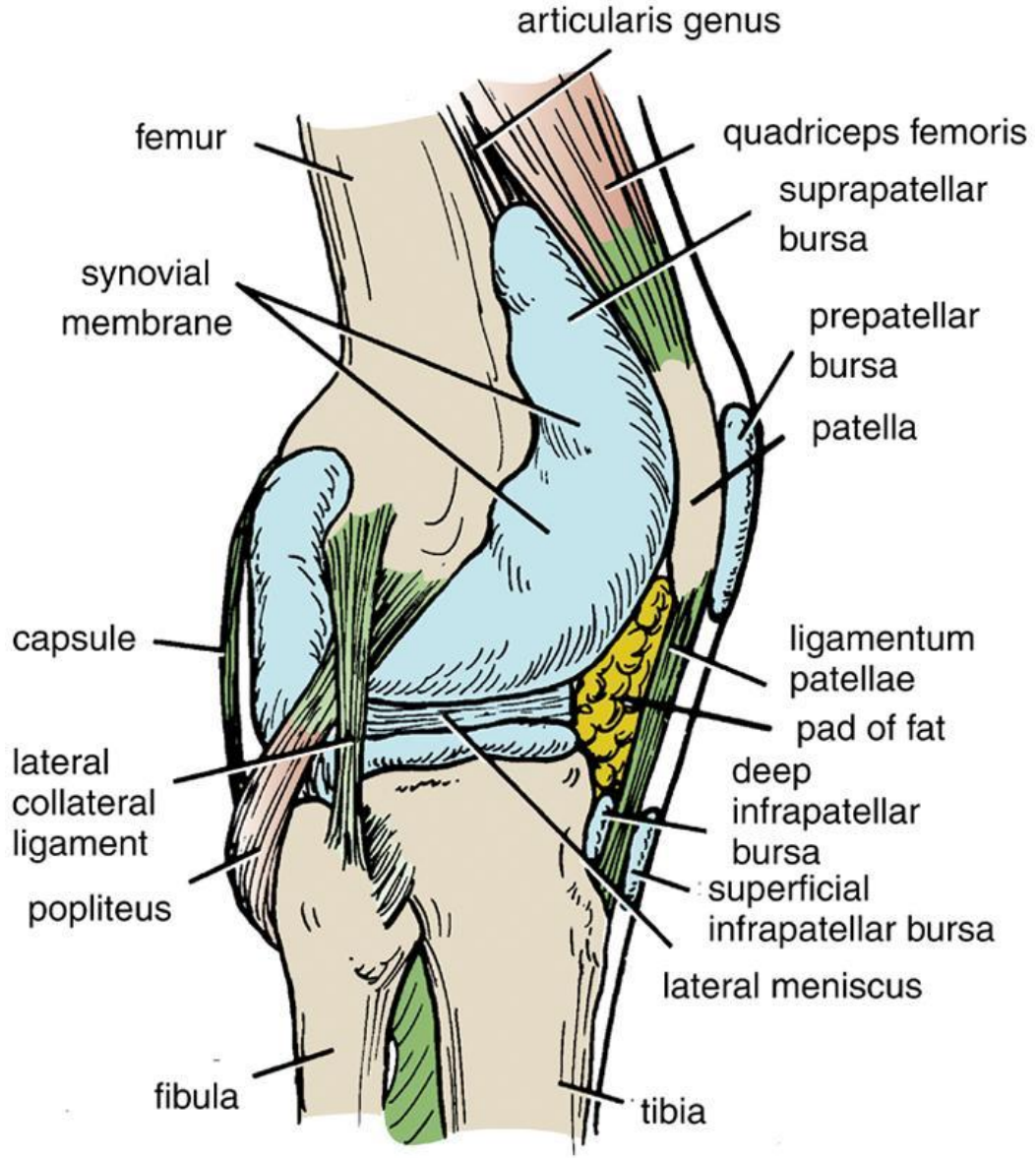
- Medial and lateral condyles of femur and Medial and lateral condyles of tibia
- Hinge joint
- Plane joint between femur and patella
- Flexion and extension and slight rotation
- Capsule is missing anteriorly
- Nerve supply: femoral, obturator, common peroneal, and tibial nerves



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Knee Joint

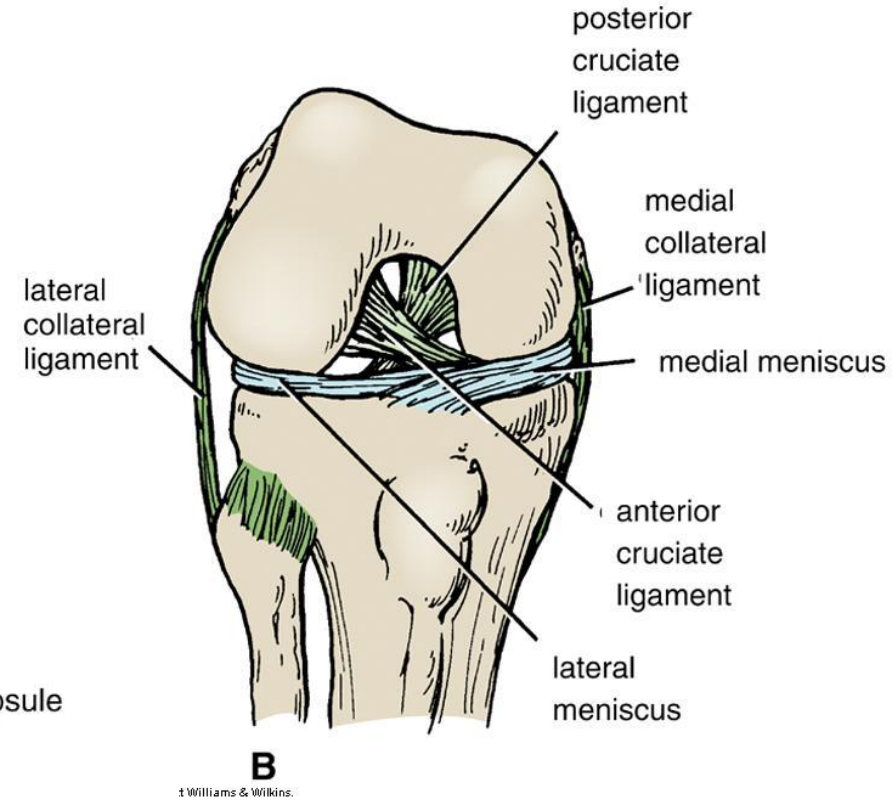
- Bursae
  - Anterior bursae
    - Suprapatellar bursa
    - Prepatellar bursa
    - Superficial infrapatellar bursa
    - Deep infrapatellar bursa
  - Posterior bursae
    - Popliteal bursa
    - Semimembranosus bursa
    - Four bursae related to muscles tendons
      - Biceps femoris
      - Sartorius and gracilis
      - Medial head of gastrocnemius
      - Lateral head of gastrocnemius

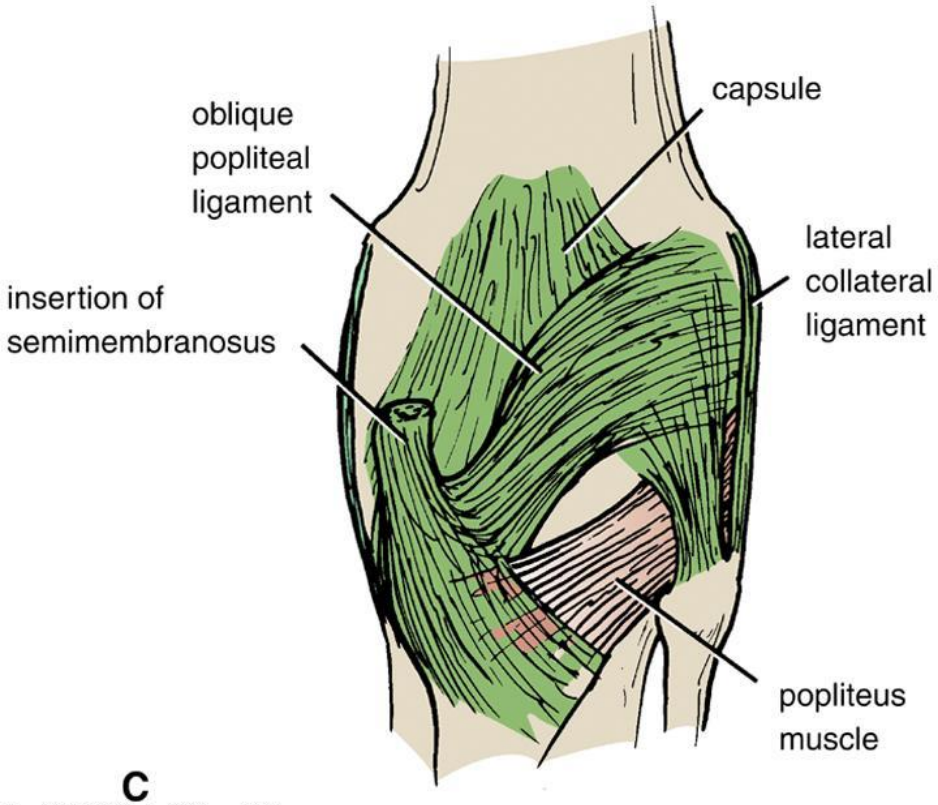


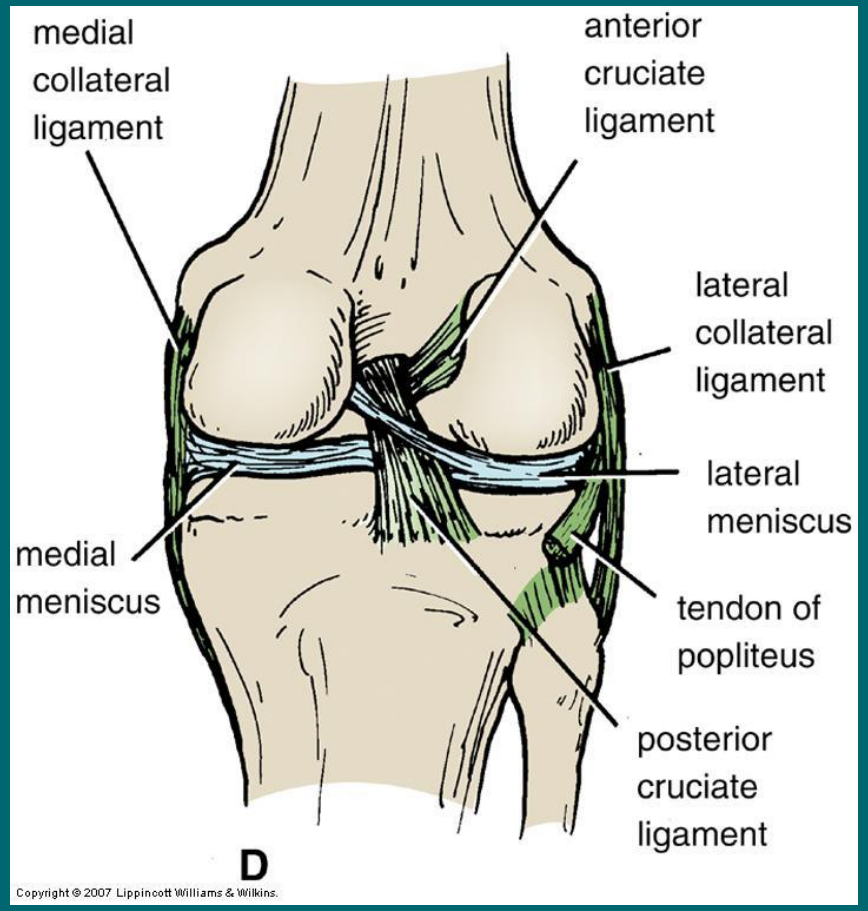
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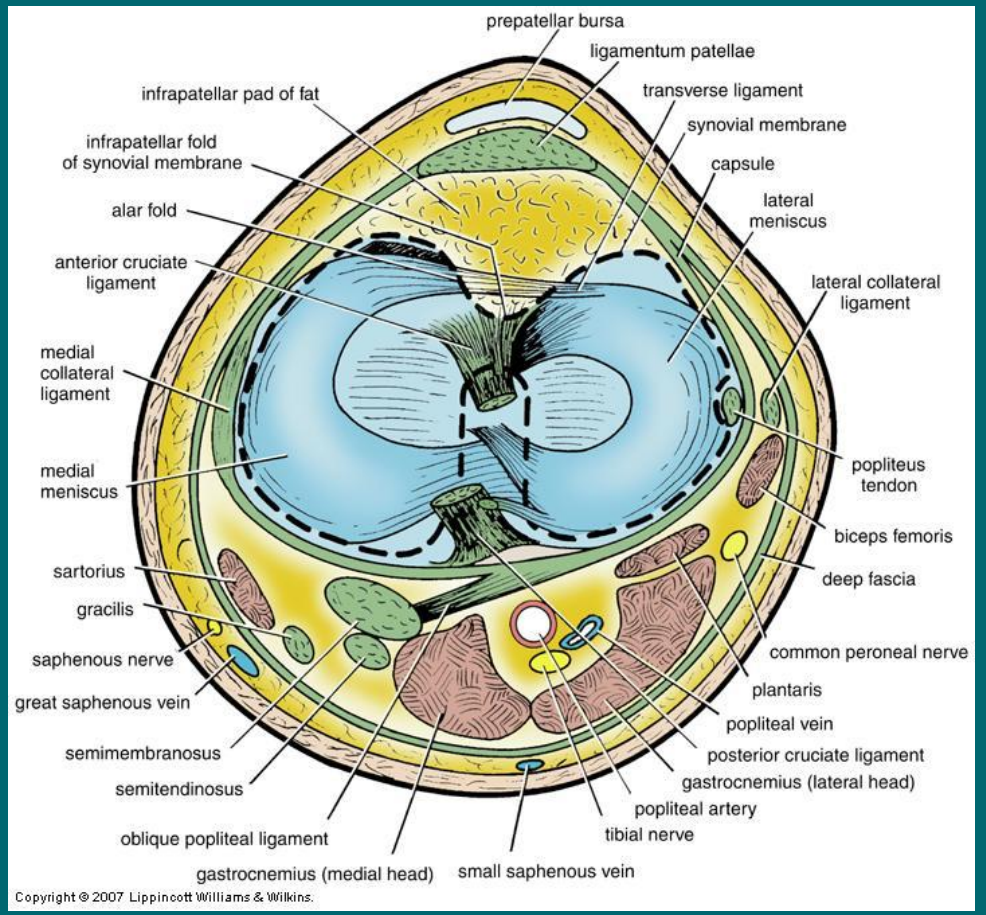
Knee Joint

- Ligaments
  - Extracapsular ligaments
    - Ligamentum patellae
    - Lateral collateral ligament
    - Medial collateral ligament
      - Attached to the medial meniscus
    - Oblique popliteal ligament
      - Expansion from the semimembranosus tendon
  - Intracapsular ligament
    - Cruciate ligaments
      - Anterior cruciate ligament
      - Posterior cruciate ligament
- Menisci
- Synovial membrane folds
  - Infrapatellar fold
  - Alar fold (free edge of the infrapatellar fold)



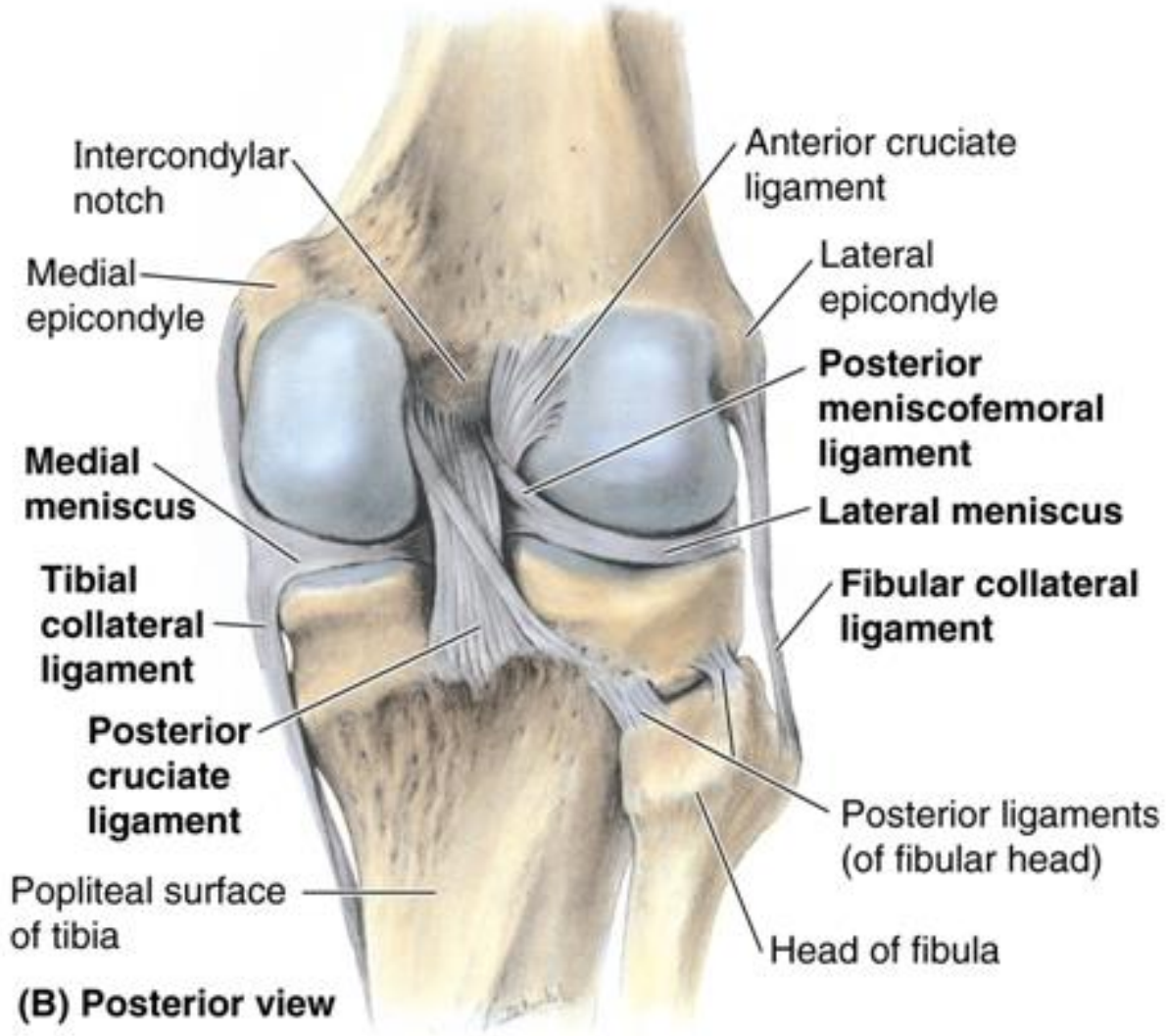



medial collateral ligament
anterior cruciate ligament
lateral collateral ligament
lateral meniscus
medial meniscus
tendon of popliteus
posterior cruciate ligament
D
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prepatellar bursa
ligamentum patellae
infrapatellar pad of fat
transverse ligament
synovial membrane
infrapatellar fold of synovial membrane
capsule
lateral meniscus
alar fold
anterior cruciate ligament
lateral collateral ligament
medial collateral ligament
popliteus tendon
medial meniscus
biceps femoris
sartorius
deep fascia
gracilis
common peroneal nerve
saphenous nerve
plantaris
great saphenous vein
popliteal vein
semimembranosus
posterior cruciate ligament
gastrocnemius (lateral head)
semitendinosus
popliteal artery
tibial nerve
oblique popliteal ligament
gastrocnemius (medial head)
small saphenous vein
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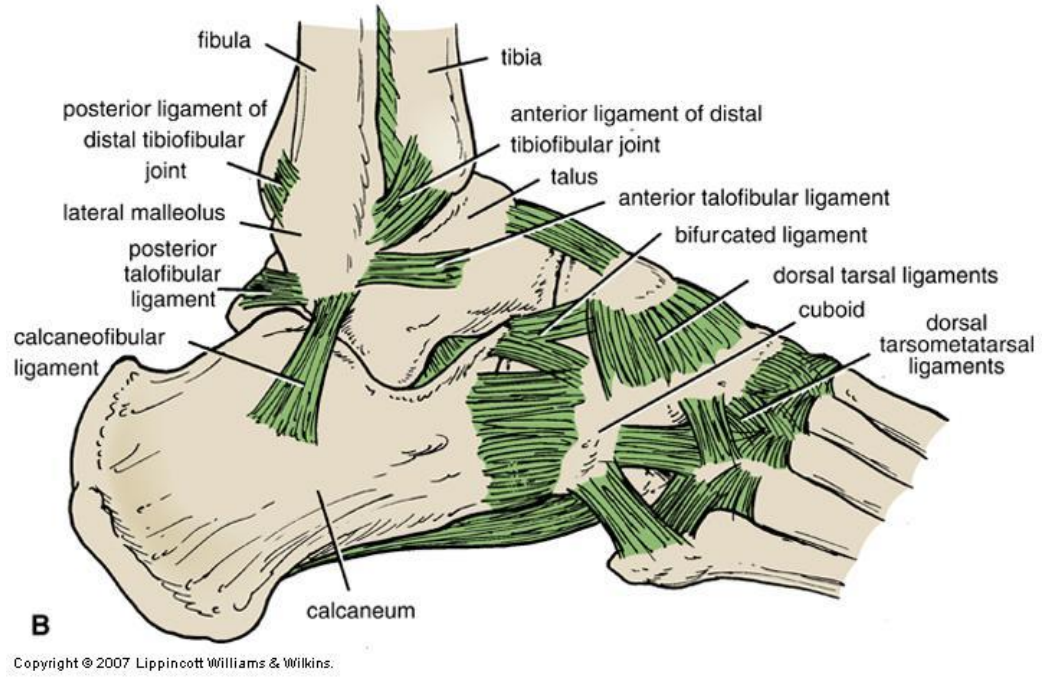
Proximal Tibiofibular Joint

- Lateral condyle of the tibia and head of fibula
- Plane joint
- Ligaments
  - Anterior and posterior ligaments
  - The interosseous membrane
- Nerve supply: common peroneal nerve

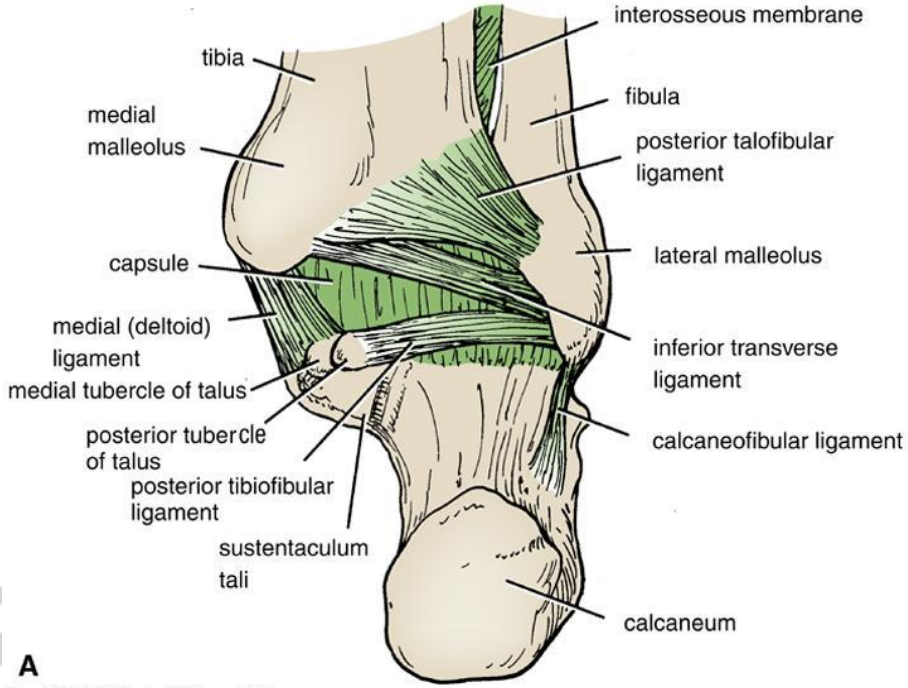


Distal Tibiofibular Joint

- Fibular notch and fibula
- Fibrous joint
- Ligaments
  - Interosseous ligaments
  - Interosseous membrane
  - Anterior and posterior ligaments
  - Inferior transverse ligament
- Nerve supply: deep peroneal and tibial nerves



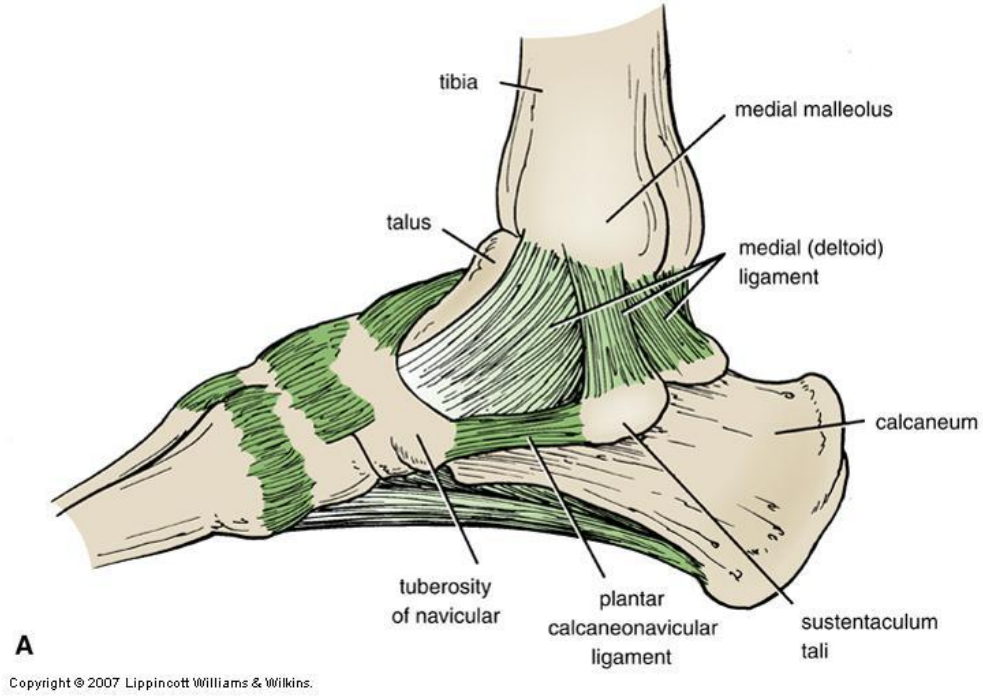
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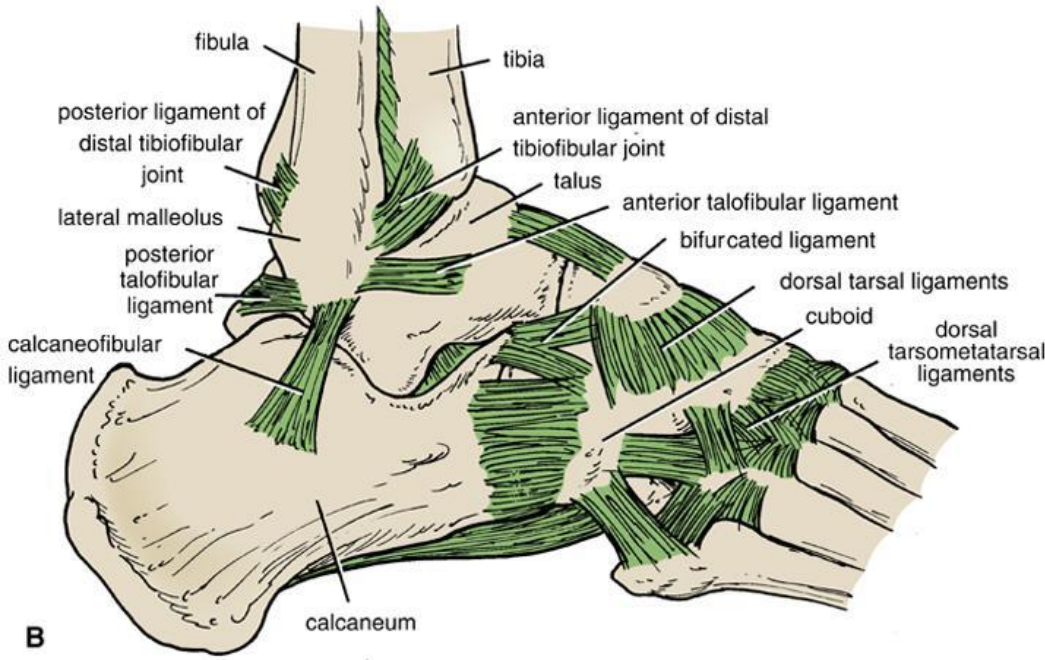
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Ankle Joint

- Distal end of tibia and fibula and the body of the talus
- Hinge joint
- Ligaments
  - Medial (deltoid) ligament
  - Lateral ligament
    - Anterior talofibular ligament
    - Calcaneofibular ligament
    - Posterior talofibular ligament
- Nerve supply: deep peroneal and tibial nerves



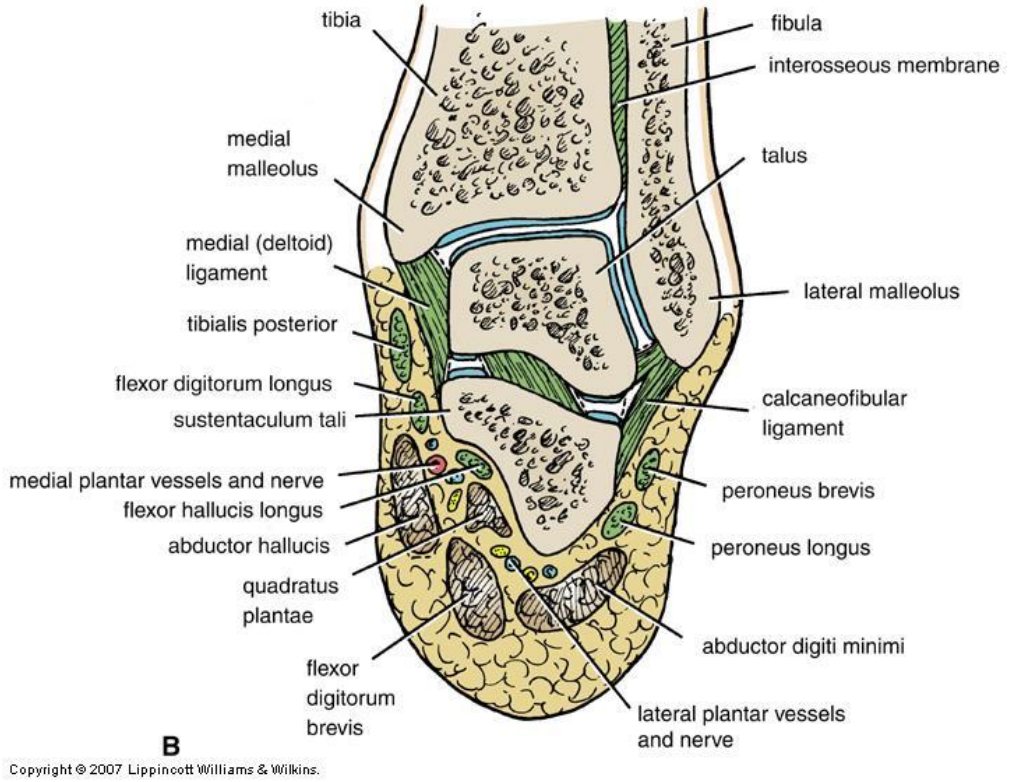
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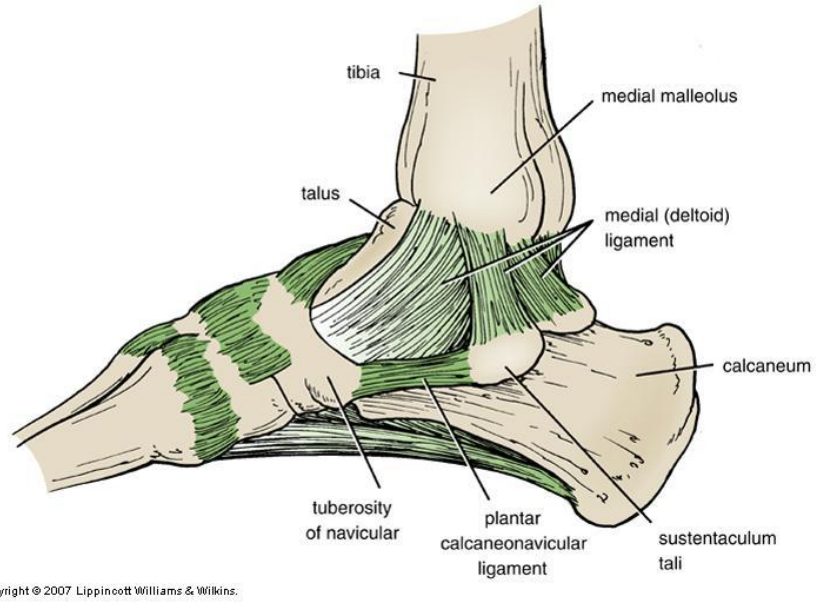


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Joints of the Foot and Toes

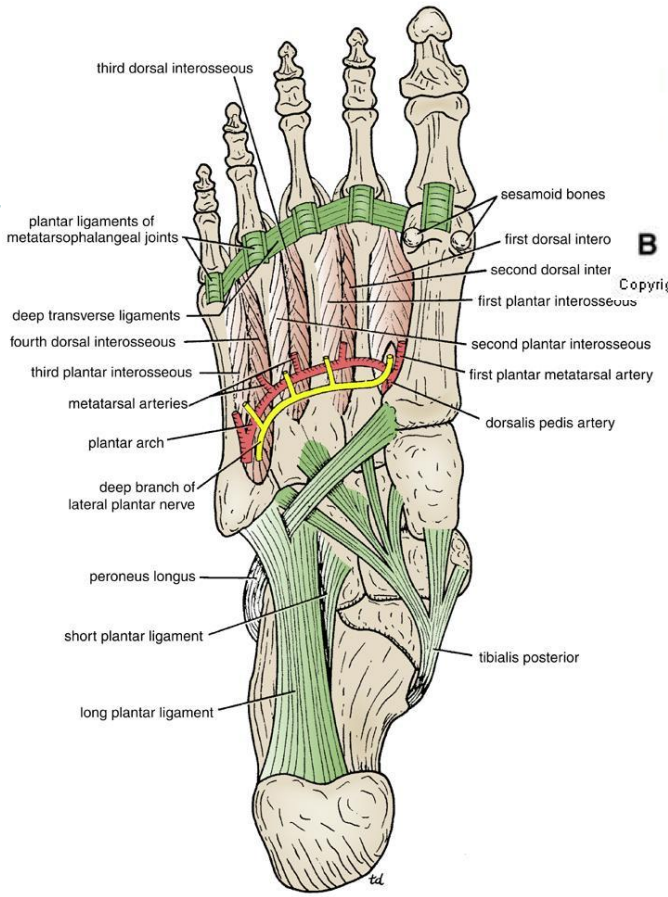
- Subtalar joint
  - Medial and lateral talocalcaneal ligaments
  - Interosseous talocalcaneal ligament
- Talocalcaneonavicular joint
  - Planter calcaneonavicular ligament
- Calcaneocuboid joint
  - Bifurcated ligament
  - Long planter ligament
  - Short planter ligament
- Talocalcaneonavicular and Calcaneocuboid joints form the midtarsal or transverse tarsal joints
- Eversion and inversion movements happen at the midtarsal and subtalar joints



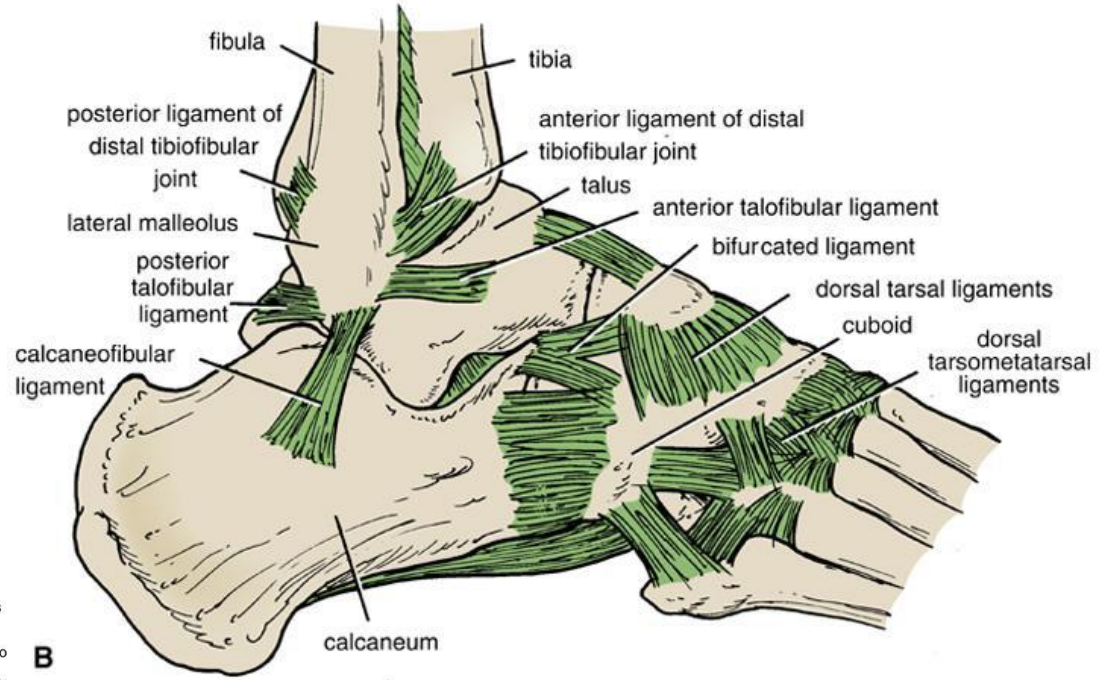


Joints of the Foot and Toes

- Cuneonavicular joint
  - Dorsal and planter ligaments
- Cuboideonavicular joint
  - Dorsal, planter and interosseous ligaments
- Intercuneiform and cuneocuboid joints
  - Dorsal, planter and interosseous ligaments
  - Same cavity with cuneonavicular joint
- Tarsometatarsal and intermetatarsal joints
  - Dorsal, planter and interosseous ligamnts
  - For big toe there is separate cavity
- Metatarsophalangeal and interphalangeal joints
  - Deep transverse ligaments



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Thanks